



New preference order relationships and their application to multiobjective interval and fuzzy interval optimization problems

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ABSTRACT

Recently, a method to generate preference ordering relationships in interval space, which is based on a family of automorphisms in bidimensional Euclidean space, and the family of φ -interval optimization problems were introduced in the literature. In this article both the method to generate preference ordering relationships in interval space and the family of φ -interval optimization problems are generalized, providing frameworks for dealing with families of multiobjective interval and multiobjective fuzzy interval optimization problems called the family of φ -multiobjective interval optimization problems and the family of φ -multiobjective fuzzy interval optimization problems, respectively. It is shown that those multiobjective interval and multiobjective fuzzy interval optimization problems presented in the literature, whose formulations are based on a given preference order relationship, can be seen as particular cases of a φ -multiobjective interval optimization problem and of a φ -multiobjective fuzzy interval optimization problems, respectively. Some optimality conditions for both these two families of optimization problems are provided and the methods, concepts and results presented herein are illustrated by several examples.

1. Introduction

Many multiobjective optimization problems, the coefficients (input data) associated with the objective functions involve constants generated by measurements, estimations, statistical analyses, and son on. Consequently, these optimization problems contain uncertain numerical data. If these problems are treated as a classical multiobjective optimization problem, then its optimal solutions may not represent the phenomena initially modeled, since no method of evaluating numerical uncertainties is adopted. In order to avoid this situation, some researchers have represented this uncertain numerical data as bounded and closed intervals of real numbers and used tools from interval analysis (see, for example, [1,8,14,15,17,18,22,26,31]).

It is well known that in order to introduce a concept of optimal solution for a multiobjective optimization problem in a determined space, it is necessary to consider a concept of order relation. Thus, it is important that the order relation adopted captures the semantic meaning of the multiobjective optimization problem, otherwise the optimal solutions may not represent the phenomena initially modeled.

Many studies about multiobjective interval optimization (see, for example, [14,15,22,26,31]) are based on one of the order relations given in [16], which belong to the family of the preference interval order relations defined in Section 4 of this paper. The

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use of preference interval order relations is appropriated to formulate a multiobjective interval optimization problem since several of these problems contain conflating objectives, that is, combination of data used to represent a given behavior of a given phenomenon. Thus, presenting a unified approach to dealing with multiobjective interval optimization problems based on preference interval order relationships is an interesting challenge. Overcoming this challenge is one of the main goals of this study. To achieve this goal, a generalization of the method to generate preference ordering relationships in interval space given recently in [9] is introduced herein. The optimization problems given by this unified approach are called φ -multiobjective interval optimization problems. Some optimality conditions for these optimization problems are presented herein.

The other goal of this study is to present a unified approach to work with multiobjective fuzzy interval optimization problems that are based on preference order relationships. This class of optimization problems whose elements are called φ -multiobjective fuzzy interval optimization problems, gives an expert the choice of the preference order relation based on the semantic meaning behind of a modeled phenomenon, thus providing a reliable optimal solution for such a multiobjective fuzzy interval optimization problems. This study also shows that a φ -multiobjective fuzzy interval optimization problem is equivalent to a classical multiobjective optimization problem. Moreover, optimality conditions for φ -multiobjective fuzzy interval optimization problems are obtained as theoretical application of the mathematical tools developed herein.

This paper is organized as follows: Section 2 presents preliminary concepts and results in interval space and in the space of all m -tuples of intervals where a generalized method to generate preference ordering relationships is developed. Then the order relationships are used to introduce the notion of φ -convexity for multi-interval-valued functions. Section 3 introduces the family of φ -multiobjective interval optimization problems and the concept of solution for these problems. Moreover, Section 3 shows that some formulations of multiobjective interval optimization problems presented in the literature can be seen as a particular optimization problem of this family and provides some optimality conditions for φ -multiobjective interval optimization problems. Section 4 presents preliminary concepts and results in the space of fuzzy intervals and in the space of the m -tuple fuzzy intervals. This section generalizes the method to generate preference ordering relationships introduced in Section 2 and it is used to introduce the notion of φ -convexity for multi-fuzzy interval-valued functions. Section 5 introduces the formulations of φ -multiobjective fuzzy interval optimization problems as well as its solution concepts. Section 5 shows that the multiobjective fuzzy interval optimization problems introduced in the literature, whose formulations are based on a given preference order relation, are particular cases of the φ -multiobjective fuzzy interval optimization problems and it also shows the φ -multiobjective fuzzy interval optimization problems are equivalent to classical multiobjective optimization problems. Moreover, Section 5 provides some optimality conditions for this family of optimization problems. Section 6 presents final considerations.

2. Interval spaces: preliminaries

This presentation uses $\mathcal{K}_C$ to denote the family $\{[a, \bar{a}] : a, \bar{a} \in \mathbb{R} \text{ and } a \leq \bar{a}\}$ of all closed and bounded intervals of real numbers equipped with the algebraic operations “+” and “ $\cdot$ ” given, respectively, by (see, e.g., [21])

$$[a, \bar{a}] + [b, \bar{b}] = [\underline{a} + \underline{b}, \bar{a} + \bar{b}] \text{ and } \lambda \cdot [a, \bar{a}] = \begin{cases} [\lambda a, \lambda \bar{a}] & \text{if } \lambda \geq 0 \\ [\lambda \bar{a}, \lambda a] & \text{if } \lambda < 0 \end{cases}.$$

The family $\mathcal{K}_C$ is called the *interval space*. It is known that $(\mathcal{K}_C, H)$ is a complete metric space (see, e.g., [2,10]), where $H : \mathcal{K}_C \times \mathcal{K}_C \rightarrow \mathbb{R}$ is the Pompeiu-Hausdorff distance given by $H([a, \bar{a}], [b, \bar{b}]) = \max\{|\underline{a} - \underline{b}|, |\bar{a} - \bar{b}|\}$ for all $[a, \bar{a}], [b, \bar{b}] \in \mathcal{K}_C$. Consequently, by defining $(\mathcal{K}_C)^m$ as being the Cartesian product composed of m -terms of $\mathcal{K}_C$, i.e.,

$$(\mathcal{K}_C)^m = \mathcal{K}_C \times \mathcal{K}_C \times \dots \times \mathcal{K}_C$$

and considering the metric $d_H : (\mathcal{K}_C)^m \times (\mathcal{K}_C)^m \rightarrow [0, \infty)$ given by

$$d_H\left(\left([a_1, \bar{a}_1], \dots, [a_m, \bar{a}_m]\right), \left([b_1, \bar{b}_1], \dots, [b_m, \bar{b}_m]\right)\right) = \max\left\{H\left([a_1, \bar{a}_1], [b_1, \bar{b}_1]\right), \dots, H\left([a_m, \bar{a}_m], [b_m, \bar{b}_m]\right)\right\},$$

it follows that $((\mathcal{K}_C)^m, d_H)$ is a complete metric space. The family $(\mathcal{K}_C)^m$ is called the *space of all m -tuples of intervals* and it can be equipped with addition “ $\boxplus$ ” and multiplication by scalar “ $\boxdot$ ”, both given via coordinate by coordinate operations. That is, given $A = (A_1, \dots, A_m)$, $B = (B_1, \dots, B_m)$, $C = (C_1, \dots, C_m)$ and $D = (D_1, \dots, D_m)$ in $(\mathcal{K}_C)^m$ and $\lambda \in \mathbb{R}$, it follows that

$$A \boxplus B = C \Leftrightarrow C_i = A_i + B_i \quad \text{and} \quad \lambda \boxdot A = D \Leftrightarrow D_i = \lambda \cdot A_i$$

for all $i \in \{1, \dots, m\}$.

Definition 2.1. Let $\varphi : \mathbb{R}^{2m} \rightarrow \mathbb{R}^{2m}$ be a bijection and let $\leq_{\mathbb{R}^{2m}}$ be an order relation on $\mathbb{R}^{2m}$. The $\varphi, \leq_{\mathbb{R}^{2m}}$ -preference order relation $\leq_{\varphi}^{2m}$ on $(\mathcal{K}_C)^m$ is defined by

$$A \leq_{\varphi}^{2m} B \Leftrightarrow \bar{\varphi}(A) \leq_{\mathbb{R}^{2m}} \bar{\varphi}(B), \quad (1)$$

where $A = ([a_1, \bar{a}_1], \dots, [a_m, \bar{a}_m])$, $B = ([b_1, \bar{b}_1], \dots, [b_m, \bar{b}_m]) \in (\mathcal{K}_C)^m$ and $\bar{\varphi} : (\mathcal{K}_C)^m \rightarrow \mathbb{R}^{2m}$ is the injective function given by

$$\bar{\varphi}([a_1, \bar{a}_1], \dots, [a_m, \bar{a}_m]) = \varphi(a_1, \bar{a}_1, \dots, a_m, \bar{a}_m).$$

It is easy to see that if $m = 1$ in Definition 2.1, then this definition coincides with Definition 3.1 in [9]. Moreover, using similar arguments to those used in the proof of Proposition 3.1 in [9], the proof of the next result is obtained.

Proposition 2.1. *The $\varphi, \leq_{\mathbb{R}^{2m}}$ -preference order relation $\leq_{\varphi}^{2m}$ on $(\mathcal{K}_C)^m$ given in Definition 2.1 is a partial order relation.*

Several phenomena can be modeled through a m -tuple of intervals whose interval components represent numerical uncertainties from different sources. For example, when buying a car, there exist different components that should be taken into account. Supposing that c_1 = purchase price from a dealer, c_2 = fuel economy, c_3 = future sale price to a dealer and c_4 = year of production are the considered components, one can use a vector (c_1, c_2, c_3, c_4) to represent all of them in an integrated way. However, this vector is composed of numerical uncertainties and not all of them have the same origin. Indeed, c_2 comes from a measurement experiment and c_3 comes from a lack of information since it is defined by sales market and this may be influenced by several factors.

It is assumed that it is possible to obtain upper and lower bounds for each one of the uncertainties in (c_1, c_2, c_3, c_4) and that each real number can be identified with a degenerated interval, it follows that the 4-tuple of intervals $(C_1, C_2, C_3, C_4) \in (\mathcal{K}_C)^4$ such that $c_i \in C_i$ for all $i \in \{1, 2, 3, 4\}$, it is a better representation for the considered components in a purchase of car since it captures the uncertainties inherent in the process. Denoting by $(C_{1A}, C_{2A}, C_{3A}, C_{4A})$ and $(C_{1B}, C_{2B}, C_{3B}, C_{4B})$ the same components mentioned above, but referring to the purchase of a car A and the purchase of a car B , respectively, it follows that to compare which purchase is the most advantageous between A and B it is necessary to consider an order relation on $(\mathcal{K}_C)^4$ that allows to compare each pair C_{iA} and C_{iB} in a different way since each one of these pairs comes from a different phenomenon. For example, while it is wished that the purchase price from a dealer be the lowest possible, it is also wished that future sale price to a dealer be the biggest possible.

Situations similar to those described above appear in many problems where decision making depends on preference order relations. In this sense, $\mathcal{A}^m$ is used to denote the particular class of automorphisms $\varphi_1 \times \varphi_2 \times \dots \times \varphi_m : \mathbb{R}^{2m} \rightarrow \mathbb{R}^{2m}$ such that

$$(\varphi_1 \times \varphi_2 \times \dots \times \varphi_m)(x_1, x_2, x_3, x_4, \dots, x_{2m-1}, x_{2m}) = \left(\varphi_1(x_1, x_2), \varphi_2(x_3, x_4), \dots, \varphi_m(x_{2m-1}, x_{2m}) \right),$$

where $\varphi_i : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ are automorphisms given by

$$\varphi_i(x_{2i-1}, x_{2i}) = (\lambda_{2i-1}x_{2i-1} + \lambda_{2i}x_{2i}, \beta_{2i-1}x_{2i-1} + \beta_{2i}x_{2i})$$

for all $(x_{2i-1}, x_{2i}) \in \mathbb{R}^2$ and for all $i \in \{1, 2, \dots, m\}$ using real numbers (that can be chosen according to a decision) $\lambda_{2i-1}, \lambda_{2i}, \beta_{2i-1}, \beta_{2i} \in \mathbb{R}$ with $\lambda_{2i-1}\beta_{2i} \neq \lambda_{2i}\beta_{2i-1}$. Thus, a particular and interesting class of $\varphi, \leq_{\mathbb{R}^{2m}}$ -preference order relations on $(\mathcal{K}_C)^m$ is that one whose elements are denoted by $\leq_{\varphi}$ with $\varphi \in \mathcal{A}^m$ and $\leq_{\mathbb{R}^{2m}}$ being the order relation $\leq^{2m}$ on $\mathbb{R}^{2m}$ given by

$$(x_1, x_2, \dots, x_{2m-1}, x_{2m}) \leq^{2m} (y_1, y_2, \dots, y_{2m-1}, y_{2m}) \Leftrightarrow (x_{2i-1}, x_{2i}) \leq (y_{2i-1}, y_{2i}) \quad (2)$$

for all $i \in \{1, 2, \dots, m\}$, where $(x_{2i-1}, x_{2i}) \leq (y_{2i-1}, y_{2i})$ if and only if $x_{2i-1} \leq y_{2i-1}$ and $x_{2i} \leq y_{2i}$ for all $i \in \{1, 2, \dots, m\}$. Therefore, in this case, the $\varphi, \leq_{\mathbb{R}^{2m}}$ -preference order relations $\leq_{\varphi}$ on $(\mathcal{K}_C)^m$, dependent on $\varphi \in \mathcal{A}^m$, are defined by

$$\begin{aligned} (A_1, \dots, A_m) \leq_{\varphi} (B_1, \dots, B_m) &\Leftrightarrow \bar{\varphi}(A_1, \dots, A_m) \leq^{2m} \bar{\varphi}(B_1, \dots, B_m) \\ &\Leftrightarrow \bar{\varphi}_i(A_i) \leq \bar{\varphi}_i(B_i) \quad \forall i \in \{1, \dots, m\} \\ &\Leftrightarrow A_i \leq_{\varphi_i} B_i \quad \forall i \in \{1, \dots, m\} \end{aligned} \quad (3)$$

for all $(A_1, \dots, A_m), (B_1, \dots, B_m) \in (\mathcal{K}_C)^m$, where $\leq_{\varphi_i}$ is a preference order relation on $\mathcal{K}_C$ given in [9] for all $i \in \{1, \dots, m\}$.

Proposition 2.2. *The preference order relation $\leq_{\varphi}$ given by (3) is a partial order relation on $(\mathcal{K}_C)^m$.*

Proof. It is well-known that $\leq$ is a partial order relation on $\mathbb{R}^2$ and from Proposition 3.1 in [9], it follows that $\leq_{\varphi_i}$ is a partial order relation on $\mathcal{K}_C$ for all $i \in \{1, \dots, m\}$. Consequently, $\leq_{\varphi}$ given by (3) is a partial order relation on $(\mathcal{K}_C)^m$. $\square$

Considering $\varphi \in \mathcal{A}^m$ and the strict partial order relations on $\mathbb{R}^{2m}$, $\leq^{2m}$ and $<^{2m}$, both given by

$$\mathbf{x} \leq^{2m} \mathbf{y} \Leftrightarrow \mathbf{x} \leq^{2m} \mathbf{y} \text{ and } \mathbf{x} \neq \mathbf{y} \quad (4)$$

and

$$\mathbf{x} <^{2m} \mathbf{y} \Leftrightarrow x_{2i-1} < y_{2i-1} \text{ and } x_{2i} < y_{2i} \text{ for all } i \in \{1, \dots, m\}, \quad (5)$$

respectively, where $\mathbf{x} = (x_1, x_2, \dots, x_{2m-1}, x_{2m}), \mathbf{y} = (y_1, y_2, \dots, y_{2m-1}, y_{2m}) \in \mathbb{R}^{2m}$, the strict partial order relations $\leq_{\varphi}$ and $<_{\varphi}$ are defined on $(\mathcal{K}_C)^m$ as follows:

$$(A_1, \dots, A_m) \leq_{\varphi} (B_1, \dots, B_m) \Leftrightarrow \bar{\varphi}(A_1, \dots, A_m) \leq^{2m} \bar{\varphi}(B_1, \dots, B_m)$$

$$\begin{aligned}
& \Leftrightarrow \begin{cases} \bar{\varphi}_i(A_i) \leq \bar{\varphi}_i(B_i) & \text{for all } i \in \{1, \dots, m\} \\ \text{and} \\ \exists k \in \{1, \dots, m\} \text{ such that } \bar{\varphi}_k(A_k) \neq \bar{\varphi}_k(B_k) \end{cases} \\
& \Leftrightarrow \begin{cases} A_i \leq_{\varphi_i} B_i & \text{for all } i \in \{1, \dots, m\} \\ \text{and} \\ \exists k \in \{1, \dots, m\} \text{ such that } A_k \neq B_k \end{cases} \\
& \Leftrightarrow \begin{cases} A_i \leq_{\varphi_i} B_i & \text{for all } i \in \{1, \dots, m\} \\ \text{and} \\ \exists k \in \{1, \dots, m\} \text{ such that } A_k \leq_{\varphi_k} B_k \end{cases} \quad (6)
\end{aligned}$$

and

$$\begin{aligned}
(A_1, \dots, A_m) <_{\varphi} (B_1, \dots, B_m) &\Leftrightarrow \bar{\varphi}(A_1, \dots, A_m) <^{2m} \bar{\varphi}(B_1, \dots, B_m) \\
&\Leftrightarrow \bar{\varphi}_i(A_i) < \bar{\varphi}_i(B_i) \quad \forall i \in \{1, \dots, m\} \\
&\Leftrightarrow A_i <_{\varphi_i} B_i \quad \forall i \in \{1, \dots, m\}, \quad (7)
\end{aligned}$$

respectively, where $\leq_{\varphi_i}$ and $<_{\varphi_i}$ are partial strict order relations on $\mathcal{K}_C$ given in [9] for all $i \in \{1, \dots, m\}$.

It is easy to see that the preference order relation given by (3), (6) and (7) are generalizations of the preference order relations given by (5), (8) and (9) in [9], respectively. The next example illustrates the applicability of this class of order relations.

Example 2.1. Given $(C_{1A}, C_{2A}, C_{3A}, C_{4A}), (C_{1B}, C_{2B}, C_{3B}, C_{4B}) \in \mathcal{K}_C^4$ the interval components referring to the purchase of a car A and a car B , respectively, both mentioned above, let C_{iA} be denoted by $[a_i, \bar{a}_i]$ and C_{iB} be denoted by $[b_i, \bar{b}_i]$ for all $i \in \{1, 2, 3, 4\}$. Moreover, given $\varphi = \varphi_1 \times \varphi_2 \times \varphi_3 \times \varphi_4 \in A^4$, let $\leq_{\varphi}$ be a preference order relation defined on $(\mathcal{K}_C)^4$. One can say that buying the car A is preferable to buying the car B (if both are comparable with respect to $\leq_{\varphi}$) whenever

$$([a_1, \bar{a}_1], [a_2, \bar{a}_2], [a_3, \bar{a}_3], [a_4, \bar{a}_4]) \leq_{\varphi} ([b_1, \bar{b}_1], [b_2, \bar{b}_2], [b_3, \bar{b}_3], [b_4, \bar{b}_4]).$$

That is, whenever $\bar{\varphi}_i([a_i, \bar{a}_i]) \leq \bar{\varphi}_i([b_i, \bar{b}_i])$ for all $i \in \{1, 2, 3, 4\}$. As criteria of comparison one can use the greatest possible purchase price and the average of the possible purchase prices for the first interval component, the variation of the possible fuel economy and the maximum of the possible fuel economy for the second interval component, the greatest possible sale price and the average of the possible sale prices for the third interval component, and the oldest and the newest year for the last interval component as follows:

$$\begin{aligned}
\varphi_1(x_1, x_2) &= (x_2, \frac{x_1 + x_2}{2}), \text{ where } \lambda_1 = 0, \lambda_2 = 1, \beta_1 = \frac{1}{2} = \beta_2; \\
\varphi_2(x_3, x_4) &= (x_4 - x_3, -x_4), \text{ where } \lambda_3 = -1, \lambda_4 = 1, \beta_3 = 0, \beta_4 = -1; \\
\varphi_3(x_5, x_6) &= (\frac{x_6 - x_5}{2}, -x_6), \text{ where } \lambda_5 = -\frac{1}{2}, \lambda_6 = \frac{1}{2}, \beta_5 = 0, \beta_6 = -1; \\
\varphi_4(x_7, x_8) &= (-x_7, -x_8), \text{ where } \lambda_7 = -1, \lambda_8 = 0, \beta_7 = 0, \beta_8 = -1.
\end{aligned}$$

Thus, $([a_1, \bar{a}_1], [a_2, \bar{a}_2], [a_3, \bar{a}_3], [a_4, \bar{a}_4]) \leq_{\varphi} ([b_1, \bar{b}_1], [b_2, \bar{b}_2], [b_3, \bar{b}_3], [b_4, \bar{b}_4])$ if and only if all the following relations hold,

$$\begin{aligned}
\bar{a}_1 &\leq \bar{b}_1 \text{ and } \frac{a_1 + \bar{a}_1}{2} \leq \frac{b_1 + \bar{b}_1}{2}, \\
\bar{a}_2 - \underline{a}_2 &\leq \bar{b}_2 - \underline{b}_2 \text{ and } -\bar{a}_2 \leq -\bar{b}_2, \\
\frac{\bar{a}_3 - \underline{a}_3}{2} &\leq \frac{\bar{b}_3 - \underline{b}_3}{2} \text{ and } -\bar{a}_3 \leq -\bar{b}_3, \\
-\underline{a}_4 &\leq -\underline{b}_4 \text{ and } -\bar{a}_4 \leq -\bar{b}_4.
\end{aligned}$$

The preference order relations given by (3), (6) and (7) are considered in the formulation of the φ -multiobjective interval optimization problems as well as its solution concepts presented in Section 3. Semantically, these preference order relations can be related to the decision maker's preferences, which makes them suitable tools to formulate the φ -multiobjective interval optimization that more closely represents the phenomenon modeled by them. To this end, consider the class of the *multi-interval-valued functions*, that is, the class of maps of type $F : S \subseteq \mathbb{R}^n \rightarrow (\mathcal{K}_C)^m$ given by $F(x) = (F_1(x), \dots, F_m(x))$, with $F_i(x) = [\underline{f}_i(x), \bar{f}_i(x)]$, where n and m are positive integers and $\underline{f}_i, \bar{f}_i : S \rightarrow \mathbb{R}$ are multivariable real-valued functions such that $\underline{f}_i(x) \leq \bar{f}_i(x)$ for all $x \in S$ and for all $i \in \{1, \dots, m\}$. If $m = 1$, then F is said to be an interval-valued function. The functions $F_1, \dots, F_n$ are called the *interval component functions* of F .

The concepts of limit and continuity for multi-interval-valued functions are naturally defined on the metric space $((\mathcal{K}_C)^m, d_H)$. Additionally, the next concepts and results are used to provide optimality conditions for the φ -multiobjective interval optimization problems.

Definition 2.2. Let $\leq_\varphi$ be the preference order relation given by (3) and let $F : S \subseteq \mathbb{R}^n \rightarrow (\mathcal{K}_C)^m$ be a multi-interval-valued function. F is said to be φ -convex if

$$F(tx + (1-t)y) \leq_\varphi t \boxplus F(x) \boxplus (1-t) \boxplus F(y) \quad (8)$$

for all $x, y \in S$ and $t \in (0, 1)$. If there exists $x^* \in S$ such that

$$F(tx^* + (1-t)y) \leq_\varphi t \boxplus F(x^*) \boxplus (1-t) \boxplus F(y) \quad (9)$$

for all $y \in S$ and $t \in (0, 1)$, then F is said to be φ -convex at x^*

Remark 2.1. Differently from some well-known concepts of convexity as h -convexity [28] and s -convexity [5], which consider the usual order relation on $\mathbb{R}$ and real-valued functions, the concept of φ -convexity given in (8) consider preference order relations on $(\mathcal{K}_C)^m$ and multi-interval-valued functions.

Proposition 2.3. Let $\leq_\varphi$ be the preference order relation given by (3) and let $F : S \subseteq \mathbb{R}^n \rightarrow (\mathcal{K}_C)^m$ be a multi-interval-valued function such that $F(x) = (F_1(x), \dots, F_m(x))$. F is φ -convex if and only if F_i is φ_i -convex for all $i \in \{1, \dots, m\}$.

Proof. From definition of $\leq_\varphi$, definitions of $\boxplus$ and $\boxminus$ and from the definition of $\bar{\varphi}$, it follows that

$$\begin{aligned} F(tx + (1-t)y) \leq_\varphi t \boxplus F(x) \boxplus (1-t) \boxplus F(y) &\Leftrightarrow \\ \left(F_1(tx + (1-t)y), \dots, F_m(tx + (1-t)y) \right) &\leq_\varphi \left(tF_1(x) + (1-t)F_1(y), \dots, tF_m(x) + (1-t)F_m(y) \right) \Leftrightarrow \\ \bar{\varphi} \left(F_1(tx + (1-t)y), \dots, F_m(tx + (1-t)y) \right) &\leq^{2m} \bar{\varphi} \left(tF_1(x) + (1-t)F_1(y), \dots, tF_m(x) + (1-t)F_m(y) \right) \Leftrightarrow \\ \bar{\varphi}_i \left(F_i(tx + (1-t)y) \right) &\leq \bar{\varphi}_i \left(tF_i(x) + (1-t)F_i(y) \right) \text{ for all } i \in \{1, \dots, m\} \Leftrightarrow \\ F_i(tx + (1-t)y) &\leq_{\varphi_i} tF_i(x) + (1-t)F_i(y) \text{ for all } i \in \{1, \dots, m\}, \end{aligned}$$

where $\bar{\varphi}_i([a, \bar{a}]) = \varphi_i(a, \bar{a})$ for all $[a, \bar{a}] \in \mathcal{K}_C$ and for all $i \in \{1, \dots, m\}$. Therefore, F is φ -convex if and only if F_i is φ_i -convex for all $i \in \{1, \dots, m\}$. $\square$

Corollary 2.1. Given $x^* \in S \subseteq \mathbb{R}^n$, let $\leq_\varphi$ be the preference order relation given by (3) and let $F : S \rightarrow (\mathcal{K}_C)^m$ be a multi-interval-valued function such that $F(x) = (F_1(x), \dots, F_m(x))$. F is φ -convex at x^* if and only if F_i is φ_i -convex at x^* for all $i \in \{1, \dots, m\}$.

Proof. This result follows directly from Definition 2.2 and from Proposition 2.3. $\square$

Remark 2.2. A real-valued function $g : S \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is said to be convex at $x^* \in S$ if

$$g(tx^* + (1-t)y) \leq tg(x^*) + (1-t)g(y)$$

for all $y \in S$ and for all $t \in (0, 1)$. Consequently, given a multi-interval-valued function $F : S \subseteq \mathbb{R}^n \rightarrow (\mathcal{K}_C)^m$, such that $F(x) = (F_1(x), \dots, F_m(x))$, with $F_i(x) = [\underline{f}_i(x), \bar{f}_i(x)]$, then from Proposition 2.3, Corollary 2.1 and from definition of the preference order given by (3), it follows that F is φ -convex at x^* if and only if the real-valued functions $[\lambda_{2i-1}\underline{f}_i + \lambda_{2i}\bar{f}_i]$ and $[\beta_{2i-1}\underline{f}_i + \beta_{2i}\bar{f}_i]$ are convex at x^* for all $i \in \{1, \dots, m\}$.

Next, some examples are given demonstrate that some concepts of convexity for multi-interval-valued functions given in the literature can be obtained from Definition 2.2.

Example 2.2. For $\varphi = (\varphi_1 \times \dots \times \varphi_m) \in \mathcal{A}^m$ such that

$$\varphi_i(x_{2i-1}, x_{2i}) = (1x_{2i-1} + 0x_{2i}, 0x_{2i-1} + 1x_{2i}) = (x_{2i-1}, x_{2i})$$

for all $i \in \{1, \dots, m\}$, consider the preference order relation $\leq_\varphi$ given by (3). Thus, the concept of φ -convexity at $x^* \in S$ for multi-interval-valued functions $F : S \subseteq \mathbb{R}^n \rightarrow (\mathcal{K}_C)^m$ coincides with the concept of LU -convexity at x^* given in [31].

Example 2.3. For $\varphi = (\varphi_1 \times \dots \times \varphi_m) \in \mathcal{A}^m$ such that

$$\varphi_i(x_{2i-1}, x_{2i}) = (1x_{2i-1} + 0x_{2i}, -1x_{2i-1} + 1x_{2i}) = (x_{2i-1}, x_{2i} - x_{2i-1})$$

for all $i \in \{1, \dots, m\}$, consider the preference order relation $\leq_\varphi$ given by (3). Thus, the concept of φ -convexity at $x^* \in S$ for multi-interval-valued functions $F : S \subseteq \mathbb{R}^n \rightarrow (\mathcal{K}_C)^m$ coincides with the concept of LS -convexity at x^* given in [26].

Example 2.4. For $\varphi = (\varphi_1 \times \dots \times \varphi_m) \in \mathcal{A}^m$ such that

$$\varphi_i(x_{2i-1}, x_{2i}) = \left(\frac{1}{2}x_{2i-1} + \frac{1}{2}x_{2i}, -\frac{1}{2}x_{2i-1} + \frac{1}{2}x_{2i}\right) = \left(\frac{x_{2i-1} + x_{2i}}{2}, \frac{x_{2i} - x_{2i-1}}{2}\right)$$

for all $i \in \{1, \dots, m\}$, consider the preference order relation $\leq_\varphi$ given by (3). Thus, the concept of φ -convexity at $x^* \in S$ for multi-interval-valued functions $F : S \subseteq \mathbb{R}^n \rightarrow (\mathcal{K}_C)^m$ coincides with the concept of CW -convexity at x^* given in [31].

3. Multiobjective interval optimization problems

We consider the following multiobjective interval minimization problem denoted by $(1MIOP_\varphi)$ and called φ -multiobjective interval minimization problem

$$(1MIOP_\varphi) \quad \min_{x \in S} F(x), \quad (10)$$

where $F : S \subseteq \mathbb{R}^n \rightarrow (\mathcal{K}_C)^m$ is a multi-interval-valued function with $F(x) = (F_1(x), \dots, F_m(x))$ and $F_i(x) = [f_i(x), \bar{f}_i(x)]$, where n and m are positive integers and $f_i, \bar{f}_i : S \rightarrow \mathbb{R}$ are multivariable real-valued functions such that $f_i(x) \leq \bar{f}_i(x)$ for all $x \in S$ and for all $i \in \{1, \dots, m\}$. Of course, an optimal solution for (10) depends on the order relation to be considered in $(\mathcal{K}_C)^m$. Hence, based on the preference ordering relation (3), both concepts of local and global solution for (10) can provide as follows. For that, denote by $N(x, \delta) := \{y \in S : \|y - x\| < \delta\}$ the δ -neighborhood of $x \in S$ with $\delta > 0$.

Definition 3.1. Let $\varphi \in \mathcal{A}^m$ be an automorphism. It is said that $\bar{x} \in S$ is a:

- (1) **(local)** strong or strict optimal solution for (10) if there does not exist $(x \in N(\bar{x}, \delta) \cap S \setminus \{\bar{x}\})$ $x \in S \setminus \{\bar{x}\}$ such that $F(x) \leq_\varphi F(\bar{x})$.
- (2) **(local)** optimal solution for (10) if there does not exist $(x \in N(\bar{x}, \delta) \cap S \setminus \{\bar{x}\})$ $x \in S \setminus \{\bar{x}\}$ such that $F(x) \leq_\varphi F(\bar{x})$.
- (3) **(local)** weak optimal solution for (10) if there does not exist $(x \in N(\bar{x}, \delta) \cap S \setminus \{\bar{x}\})$ $x \in S \setminus \{\bar{x}\}$ such that $F(x) <_\varphi F(\bar{x})$.

The case of multiobjective interval maximization problem denoted by $(2MIOP_\varphi)$ and called φ -multiobjective interval maximization problem, is obtained in the same way.

$$(2MIOP_\varphi) \quad \max_{x \in S} F(x), \quad (11)$$

which results in the following definition.

Definition 3.2. Let $\varphi \in \mathcal{A}^m$ be an automorphism. It is said that $\bar{x} \in S$ is a:

- (1) **(local)** strong or strict optimal solution for (11) if there does not exist $(x \in N(\bar{x}, \delta) \cap S \setminus \{\bar{x}\})$ $x \in S \setminus \{\bar{x}\}$ such that $F(\bar{x}) \leq_\varphi F(x)$.
- (2) **(local)** optimal solution for (11) if there does not exist $(x \in N(\bar{x}, \delta) \cap S \setminus \{\bar{x}\})$ $x \in S \setminus \{\bar{x}\}$ such that $F(\bar{x}) \leq_\varphi F(x)$.
- (3) **(local)** weak optimal solution for (11) if there does not exist $(x \in N(\bar{x}, \delta) \cap S \setminus \{\bar{x}\})$ $x \in S \setminus \{\bar{x}\}$ such that $F(\bar{x}) <_\varphi F(x)$.

The following relations are immediate:

strong or strict optimal solution $\Rightarrow$ optimal solution $\Rightarrow$ weak optimal solution

Behind each of the problems $(1MIOP_\varphi)$ and $(2MIOP_\varphi)$ there exists a semantic that is expressed by means of its optimal solution concepts. The optimal solution concept for each one of these problems depends on the $\varphi \in \mathcal{A}^m$ that is defined through the values $\lambda_{2i-1}, \lambda_{2i}, \beta_{2i-1}, \beta_{2i} \in \mathbb{R}$ with $\lambda_{2i-1}\beta_{2i} \neq \lambda_{2i}\beta_{2i-1}$ for all $i \in \{1, \dots, m\}$. Consequently, the optimal solution concepts for these problems depend on the values $\lambda_{2i-1}, \lambda_{2i}, \beta_{2i-1}, \beta_{2i} \in \mathbb{R}$. Therefore, the semantics related to $(1MIOP_\varphi)$ and $(2MIOP_\varphi)$ are implicitly expressed by the values $\lambda_{2i-1}, \lambda_{2i}, \beta_{2i-1}, \beta_{2i} \in \mathbb{R}$ which can be chosen by a decision maker.

Some researchers studied (10) and (11) considering some particular order relations. Next, some formulations of multiobjective interval optimization problems as well as its concepts of optimal solution which have appeared in the literature are presented as particular cases of our formulation.

Example 3.1. Consider $S \subseteq \mathbb{R}^n$ and $\varphi = (\varphi_1 \times \dots \times \varphi_m) \in \mathcal{A}^m$ given in Example 2.2. Then $(1MIOP_\varphi)$ coincides with the formulation of the multiobjective interval minimization problem given in [31]. The concept of solution for $(1MIOP_\varphi)$ coincides with the concept of solution given by Definition 3.1 in [31].

Example 3.2. Consider $S \subseteq \mathbb{R}^n$ and $\varphi = (\varphi_1 \times \dots \times \varphi_m) \in \mathcal{A}^m$ given in Example 2.3. Then $(1MIOP_\varphi)$ coincides with the formulation of the multiobjective interval minimization problem given in [26]. The concept of solution for $(1MIOP_\varphi)$ coincides with the concept of solution given by Definition 3.4 in [26].

Example 3.3. Consider $S \subseteq \mathbb{R}^n$ and $\varphi = (\varphi_1 \times \dots \times \varphi_m) \in \mathcal{A}^m$ given in Example 2.4. Then $(1MIOP_\varphi)$ coincides with the formulation of the multiobjective interval minimization problem given in [31]. The concept of solution for $(1MIOP_\varphi)$ coincides with the concept of solution given by Definition 3.2 in [31].

3.1. Optimality conditions for multiobjective interval optimization problems with preference ordering relation

Some optimality conditions for multiobjective interval optimization problems presented in the literature, where the use of preference order relations is adopted, can be obtained using the mathematical tools developed herein as illustrated below.

Example 3.4 (Part I of Theorem 4.2 in [31]). Given $\mathbf{0} = (0, \dots, 0) \in \mathbb{R}^n$, consider $(1MIOP_\varphi)$ such that φ is the automorphism given in Example 2.2, $S = \{x \in \mathbb{R}^n : g_j(x) \leq 0 \text{ for all } j \in \{1, \dots, k\}\}$ with $g_j : \mathbb{R}^n \rightarrow \mathbb{R}$ convex on $\mathbb{R}^n$ and differentiable at $\bar{x}$ for all $j \in \{1, \dots, k\}$, $\bar{x} \in S$ and $F : S \rightarrow (\mathcal{K}_C)^m$ given by $F(x) = (F_1(x), \dots, F_m(x))$ with $F_i(x) = [\underline{f}_i(x), \bar{f}_i(x)]$, where $\underline{f}_i, \bar{f}_i : S \rightarrow \mathbb{R}$ are continuously differentiable at $\bar{x}$ for all $i \in \{1, \dots, m\}$. If F is φ -convex at $\bar{x}$ and if there exist $w_1, \dots, w_{2m} \in \mathbb{R}$ with $w_i > 0$ for all $i \in \{1, \dots, 2m\}$ and $\mu_1, \dots, \mu_k \in \mathbb{R}$ with $\mu_j \geq 0$ for all $j \in \{1, \dots, k\}$ such that

$$\begin{aligned} \text{(i)} \quad & \sum_{i=1}^m w_{2i-1} \nabla \underline{f}_i(\bar{x}) + \sum_{i=1}^m w_{2i} \nabla \bar{f}_i(\bar{x}) + \sum_{j=1}^k \mu_j \nabla g_j(\bar{x}) = \mathbf{0}; \\ \text{(ii)} \quad & \mu_j g_j(\bar{x}) = 0 \text{ for all } j \in \{1, \dots, k\}, \end{aligned}$$

then $\bar{x}$ is an optimal solution for $(1MIOP_\varphi)$.

Example 3.5 (Part II of Theorem 4.2 in [31]). Given $\mathbf{0} = (0, \dots, 0) \in \mathbb{R}^n$, consider $(1MIOP_\varphi)$ such that φ is the automorphism given in Example 2.4, $S = \{x \in \mathbb{R}^n : g_j(x) \leq 0 \text{ for all } j \in \{1, \dots, k\}\}$ with $g_j : \mathbb{R}^n \rightarrow \mathbb{R}$ convex on $\mathbb{R}^n$ and differentiable at $\bar{x}$ for all $j \in \{1, \dots, k\}$, $\bar{x} \in S$ and $F : S \rightarrow (\mathcal{K}_C)^m$ given by $F(x) = (F_1(x), \dots, F_m(x))$ with $F_i(x) = [\underline{f}_i(x), \bar{f}_i(x)]$, where $\underline{f}_i, \bar{f}_i : S \rightarrow \mathbb{R}$ are continuously differentiable at $\bar{x}$ for all $i \in \{1, \dots, m\}$. If F is φ -convex at $\bar{x}$ and if there exist $w_1, \dots, w_{2m} \in \mathbb{R}$ with $w_i > 0$ for all $i \in \{1, \dots, 2m\}$ and $\mu_1, \dots, \mu_k \in \mathbb{R}$ with $\mu_j \geq 0$ for all $j \in \{1, \dots, k\}$ such that

$$\begin{aligned} \text{(i)} \quad & \sum_{i=1}^m w_{2i-1} \nabla \left(\frac{\underline{f}_i(\bar{x}) + \bar{f}_i(\bar{x})}{2} \right) + \sum_{i=1}^m w_{2i} \nabla \left(\frac{\bar{f}_i(\bar{x}) - \underline{f}_i(\bar{x})}{2} \right) + \sum_{j=1}^k \mu_j \nabla g_j(\bar{x}) = \mathbf{0}; \\ \text{(ii)} \quad & \mu_j g_j(\bar{x}) = 0 \text{ for all } j \in \{1, \dots, k\}, \end{aligned}$$

then $\bar{x}$ is an optimal solution for $(1MIOP_\varphi)$.

Example 3.6 (Part III of Theorem 4.2 in [31]). Given $\mathbf{0} = (0, \dots, 0) \in \mathbb{R}^n$, consider $(1MIOP_\varphi)$ such that φ is the automorphism given in Example 2.4,

$$S = \{x \in \mathbb{R}^n : g_j(x) \leq 0 \text{ for all } j \in \{1, \dots, k\}\}$$

with $g_j : \mathbb{R}^n \rightarrow \mathbb{R}$ convex on $\mathbb{R}^n$ and differentiable at $\bar{x}$ for all $j \in \{1, \dots, k\}$, $\bar{x} \in S$ and $F : S \rightarrow (\mathcal{K}_C)^m$ given by $F(x) = (F_1(x), \dots, F_m(x))$ with $F_i(x) = [\underline{f}_i(x), \bar{f}_i(x)]$, where $\underline{f}_i, \bar{f}_i : S \rightarrow \mathbb{R}$ are continuously differentiable at $\bar{x}$ for all $i \in \{1, \dots, m\}$. If F is φ -convex at $\bar{x}$ and if there exist $w_1, \dots, w_{2m} \in \mathbb{R}$ with $w_i > 0$ for all $i \in \{1, \dots, 2m\}$ and $\mu_1, \dots, \mu_k \in \mathbb{R}$ with $\mu_j \geq 0$ for all $j \in \{1, \dots, k\}$ such that

$$\begin{aligned} \text{(i)} \quad & w_{2i-1} < w_{2i} \text{ for all } i \in \{1, \dots, m\} \\ \text{(ii)} \quad & \sum_{i=1}^m w_{2i-1} \nabla \underline{f}_i(\bar{x}) + \sum_{i=1}^m w_{2i} \nabla \bar{f}_i(\bar{x}) + \sum_{j=1}^k \mu_j \nabla g_j(\bar{x}) = \mathbf{0}; \\ \text{(iii)} \quad & \mu_j g_j(\bar{x}) = 0 \text{ for all } j \in \{1, \dots, k\}, \end{aligned}$$

then $\bar{x}$ is an optimal solution for $(1MIOP_\varphi)$.

Example 3.7 (Theorem 4.3 in [26]). Given $\mathbf{0} = (0, \dots, 0) \in \mathbb{R}^n$, consider $(1MIOP_\varphi)$ such that φ is the automorphism given in Example 2.3, $S = \{x \in \mathbb{R}^n : g_j(x) \leq 0 \text{ for all } j \in \{1, \dots, k\}\}$ with $g_j : \mathbb{R}^n \rightarrow \mathbb{R}$ convex on $\mathbb{R}^n$ and differentiable at $\bar{x}$ for all $j \in \{1, \dots, k\}$, $\bar{x} \in S$ and $F : S \rightarrow (\mathcal{K}_C)^m$ given by $F(x) = (F_1(x), \dots, F_m(x))$ with $F_i(x) = [\underline{f}_i(x), \bar{f}_i(x)]$, where $\underline{f}_i, \bar{f}_i : S \rightarrow \mathbb{R}$ are continuously

differentiable at $\bar{x}$ for all $i \in \{1, \dots, m\}$. If F is φ -convex at $\bar{x}$ and if there exist $w_1, \dots, w_{2m} \in \mathbb{R}$ with $w_i > 0$ for all $i \in \{1, \dots, 2m\}$ and $\mu_1, \dots, \mu_k \in \mathbb{R}$ with $\mu_j \geq 0$ for all $j \in \{1, \dots, k\}$ such that

$$(i) \sum_{i=1}^m w_{2i-1} \nabla f_i(\bar{x}) + \sum_{i=1}^m w_{2i} \nabla (\bar{f}_i(\bar{x}) - \underline{f}_i(\bar{x})) + \sum_{j=1}^k \mu_j \nabla g_j(\bar{x}) = 0;$$

$$(ii) \mu_j g_j(\bar{x}) = 0 \text{ for all } j \in \{1, \dots, k\},$$

then $\bar{x}$ is an optimal solution for $(1MIO P_\varphi)$.

Next is shown that solving $(1MIO P_\varphi)$ and $(2MIO P_\varphi)$ is equivalent to solving a classical multiobjective optimization problem of type

$$\min_{x \in S} (h_1(x), h_2(x), \dots, h_{2m}(x)) \quad (12)$$

where $S \subseteq \mathbb{R}^n$ and $h_1, h_2, \dots, h_{2m} : S \rightarrow \mathbb{R}$. Recall that $\bar{x} \in S$ is an efficient solution for (12) if there does not exist $x \in S \setminus \{\bar{x}\}$ such that $h_i(x) \leq h_i(\bar{x})$ for all $i \in \{1, \dots, 2m\}$, and $\bar{x} \in S$ is a weak efficient solution for (12) if there does not exist $x \in S \setminus \{\bar{x}\}$ such that $h_i(x) < h_i(\bar{x})$ for all $i \in \{1, \dots, 2m\}$.

The composition $\bar{\varphi} \circ F : S \rightarrow \mathbb{R}^{2m}$ defined by

$$\begin{aligned} (\bar{\varphi} \circ F)(x) &= \bar{\varphi}(F(x)) = \bar{\varphi}(F_1(x), \dots, F_m(x)) = \bar{\varphi}([f_1(x), \bar{f}_1(x)], \dots, [f_m(x), \bar{f}_m(x)]) \\ &= \left(\varphi_1(\underline{f}_1(x), \bar{f}_1(x)), \dots, \varphi_m(\underline{f}_m(x), \bar{f}_m(x)) \right) \\ &= \left((\lambda_1 \underline{f}_1 + \lambda_2 \bar{f}_1)(x), (\beta_1 \underline{f}_1 + \beta_2 \bar{f}_1)(x), \dots, (\lambda_{2m-1} \underline{f}_m + \lambda_{2m} \bar{f}_m)(x), (\beta_{2m-1} \underline{f}_m + \beta_{2m} \bar{f}_m)(x) \right) \\ &= \left(\Lambda_1^T f(x), B_1^T f(x), \dots, \Lambda_{2m}^T f(x), B_{2m}^T f(x) \right), \end{aligned}$$

where $\Lambda_1 = (\lambda_1, \lambda_2, 0, \dots, 0, 0)$, $\Lambda_2 = (0, 0, \lambda_3, \lambda_4, 0, \dots, 0)$, $\Lambda_i = (0, 0, \dots, \lambda_{2i-1}, \lambda_{2i}, 0, \dots, 0)$, $\dots$, $\Lambda_m = (0, 0, \dots, \lambda_{2m-1}, \lambda_{2m})$, $B_1 = (\beta_1, \beta_2, 0, \dots, 0, 0)$, $B_2 = (0, 0, \beta_3, \beta_4, 0, \dots, 0)$, $B_i = (0, 0, \dots, \beta_{2i-1}, \beta_{2i}, 0, \dots, 0)$, $\dots$, $B_m = (0, 0, \dots, \beta_{2m-1}, \beta_{2m}) \in \mathbb{R}^{2m}$ and $f(x) = (\underline{f}_1(x), \bar{f}_1(x), \underline{f}_2(x), \bar{f}_2(x), \dots, \underline{f}_m(x), \bar{f}_m(x))$ for all $x \in S$, is a vector function since $F : S \subseteq \mathbb{R}^n \rightarrow (\mathcal{K}_C)^m$ is a multi-interval-valued function and $\bar{\varphi} : (\mathcal{K}_C)^m \rightarrow \mathbb{R}^{2m}$ is given by the automorphism $\varphi : \mathbb{R}^{2m} \rightarrow \mathbb{R}^{2m}$, $\varphi \in \mathcal{A}^m$. Thus, associated with the φ -multiobjective interval optimization problems (10) and (11), consider the following classical multiobjective minimization problem.

$$(MOP_\varphi) \min_{x \in S} (\bar{\varphi} \circ F)(x). \quad (13)$$

Using the usual partial order relation (2) in $\mathbb{R}^{2m}$ and strict partial order relations (4) and (5), the usual solution (efficient and weakly efficient) concept for (13) is obtained.

The next results show the equivalence between the problems $(1MIO P_\varphi)$, $(2MIO P_\varphi)$ and the multiobjective optimization problem (13).

Theorem 3.1. $\bar{x} \in S$ is an optimal solution (weakly optimal solution) for $(1MIO P_\varphi)$ if and only if $\bar{x} \in S$ is an efficient solution (weakly efficient solution) for (MOP_φ) .

Proof. Let $\bar{x} \in S$ be an optimal solution for $(1MIO P_\varphi)$. If $\bar{x}$ is not an efficient solution for (MOP_φ) , then there exists $x \in S$, with $x \neq \bar{x}$, such that

$$\Lambda_i^T f(x) \leq \Lambda_i^T f(\bar{x}) \quad \text{and} \quad B_i^T f(x) \leq B_i^T f(\bar{x})$$

for all $i \in \{1, \dots, m\}$ and, consequently,

$$\begin{aligned} (\varphi_1(\underline{f}_1(x), \bar{f}_1(x)), \dots, \varphi_m(\underline{f}_m(x), \bar{f}_m(x))) &\leq^{2m} (\varphi_1(\underline{f}_1(\bar{x}), \bar{f}_1(\bar{x})), \dots, \varphi_m(\underline{f}_m(\bar{x}), \bar{f}_m(\bar{x}))) \Rightarrow \\ &\Rightarrow \varphi(\underline{f}_1(x), \bar{f}_1(x), \dots, \underline{f}_m(x), \bar{f}_m(x)) \leq^{2m} \varphi(\underline{f}_1(\bar{x}), \bar{f}_1(\bar{x}), \dots, \underline{f}_m(\bar{x}), \bar{f}_m(\bar{x})) \Rightarrow \\ &\Rightarrow \bar{\varphi}([f_1(x), \bar{f}_1(x)], \dots, [f_m(x), \bar{f}_m(x)]) \leq^{2m} \bar{\varphi}([f_1(\bar{x}), \bar{f}_1(\bar{x})], \dots, [f_m(\bar{x}), \bar{f}_m(\bar{x})]) \Rightarrow \\ &\Rightarrow \bar{\varphi}(F(x)) \leq^{2m} \bar{\varphi}(F(\bar{x})) \Rightarrow F(x) \leq_\varphi F(\bar{x}). \end{aligned}$$

However, this contradicts the fact that $\bar{x}$ is an optimal solution for $(1MIO P_\varphi)$. Therefore, if $\bar{x} \in S$ is an optimal solution for $(1MIO P_\varphi)$, then $\bar{x} \in S$ is an efficient solution for (MOP_φ) .

Reciprocally, let $\bar{x} \in S$ be an efficient solution for (MOP_φ) . If $\bar{x}$ is not an optimal solution for $(1MIO P_\varphi)$, then there exists $x \in S$, with $x \neq \bar{x}$, such that $F(x) \leq_\varphi F(\bar{x})$, that is, $\bar{\varphi}(F(x)) \leq^{2m} \bar{\varphi}(F(\bar{x}))$. Thus,

$$\Lambda_i^T f(x) \leq \Lambda_i^T f(\bar{x}) \quad \text{and} \quad B_i^T f(x) \leq B_i^T f(\bar{x}),$$

for all $i \in \{1, \dots, m\}$. However, this contradicts the fact that $\bar{x}$ is an efficient solution for (MOP_φ) . Therefore, if $\bar{x} \in S$ is an efficient solution for (MOP_φ) , then $\bar{x} \in S$ is an optimal solution for $(1MIO P_\varphi)$.

Using similar arguments, the proof for weak solutions can be obtained. $\square$

Theorem 3.2. $\bar{x} \in S$ is an optimal solution (weak optimal solution) for $(2MIO P_\varphi)$ if and only if $\bar{x} \in S$ is an efficient solution (weakly efficient solution) for $(MOP_{(-\varphi)})$.

Proof. Given $-\varphi \in \mathcal{A}^m$ by

$$-\varphi(a_1, \dots, a_{2m}) = ((-\lambda_1 a_1 - \lambda_2 a_2, -\beta_1 a_1 - \beta_2 a_2), \dots, (-\lambda_{2m-1} a_{2m-1} - \lambda_{2m} a_{2m}, -\beta_{2m-1} a_{2m-1} - \beta_{2m} a_{2m})),$$

consider

$$(MOP_{(-\varphi)}) \min_{x \in S} (-\Lambda_1^T f(x), -B_1^T f(x), \dots, -\Lambda_m^T f(x), -B_m^T f(x)).$$

Let $\bar{x} \in S$ be an optimal solution for $(2MIO P_\varphi)$. If $\bar{x}$ is not an efficient solution for $(MOP_{(-\varphi)})$, then there exists $x \in S$, with $x \neq \bar{x}$, such that

$$-\Lambda_i^T f(x) \leq -\Lambda_i^T f(\bar{x}) \quad \text{and} \quad -B_i^T f(x) \leq -B_i^T f(\bar{x}),$$

for all $i \in \{1, \dots, m\}$ and, consequently,

$$\begin{aligned} (\varphi_1(\underline{f}_1(\bar{x}), \bar{f}_1(\bar{x})), \dots, \varphi_m(\underline{f}_m(\bar{x}), \bar{f}_m(\bar{x}))) &\leq^{2m} (\varphi_1(\underline{f}_1(x), \bar{f}_1(x)), \dots, \varphi_m(\underline{f}_m(x), \bar{f}_m(x))) \Rightarrow \\ &\Rightarrow \varphi(\underline{f}_1(\bar{x}), \bar{f}_1(\bar{x}), \dots, \underline{f}_m(\bar{x}), \bar{f}_m(\bar{x})) \leq^{2m} \varphi(\underline{f}_1(x), \bar{f}_1(x), \dots, \underline{f}_m(x), \bar{f}_m(x)) \Rightarrow \\ &\Rightarrow \bar{\varphi}([\underline{f}_1(\bar{x}), \bar{f}_1(\bar{x})], \dots, [\underline{f}_m(\bar{x}), \bar{f}_m(\bar{x})]) \leq^{2m} \bar{\varphi}([\underline{f}_1(x), \bar{f}_1(x)], \dots, [\underline{f}_m(x), \bar{f}_m(x)]) \Rightarrow \\ &\Rightarrow \bar{\varphi}(F(\bar{x})) \leq^{2m} \bar{\varphi}(F(x)) \Rightarrow F(\bar{x}) \leq_\varphi F(x). \end{aligned}$$

However, this contradicts the fact that $\bar{x}$ is an optimal solution for $(2MIO P_\varphi)$. Therefore, if $\bar{x} \in S$ is an optimal solution for $(2MIO P_\varphi)$, then $\bar{x}$ is an efficient solution for $(MOP_{(-\varphi)})$.

Reciprocally, let $\bar{x} \in S$ be an efficient solution for $(MOP_{(-\varphi)})$. If $\bar{x}$ is not an optimal solution for $(2MIO P_\varphi)$, then there exists $x \in S$, with $x \neq \bar{x}$, such that $F(\bar{x}) \leq_\varphi F(x)$ and, consequently,

$$\bar{\varphi}(F(\bar{x})) \leq_\varphi \bar{\varphi}(F(x)) \Rightarrow \varphi(\underline{f}_1(\bar{x}), \bar{f}_1(\bar{x}), \dots, \underline{f}_m(\bar{x}), \bar{f}_m(\bar{x})) \leq^{2m} \varphi(\underline{f}_1(x), \bar{f}_1(x), \dots, \underline{f}_m(x), \bar{f}_m(x)).$$

Thus,

$$\Lambda_i^T f(\bar{x}) \leq \Lambda_i^T f(x) \quad \text{and} \quad B_i^T f(\bar{x}) \leq B_i^T f(x)$$

that is,

$$-\Lambda_i^T f(x) \leq -\Lambda_i^T f(\bar{x}) \quad \text{and} \quad -B_i^T f(x) \leq -B_i^T f(\bar{x})$$

for all $i \in \{1, \dots, m\}$. However, this contradicts the fact that $\bar{x}$ is an efficient solution for $(MOP_{(-\varphi)})$. Therefore, if $\bar{x} \in S$ is an efficient solution for $(MOP_{(-\varphi)})$, then $\bar{x}$ is an optimal solution for $(2MIO P_\varphi)$.

Using similar arguments, the proof for weak solutions can be obtained. $\square$

Theorems 3.1 and 3.2 demonstrate that to solve the problems (10) and (11), the usual multiobjective programming techniques can be used such as scalarization, interactive methods and optimality condition of type Karush-Kuhn-Tucker (see [7], [20], [25]). This which allows for the unification of the optimal conditions for multiobjective interval optimization problems given through preference order relations, such as Theorem 4.2 in [31] and Theorem 4.3 in [26] illustrated in Examples 3.4, 3.5, 3.6 and 3.7. The following result helps to present some optimality conditions for φ -multiobjective interval optimization problems in a concise way.

Theorem 3.3. Let $F : S \subseteq \mathbb{R}^n \rightarrow (\mathcal{K}_C)^m$ be a multi-interval-valued function given by $F(x) = (F_1(x), \dots, F_m(x))$, with $F_i(x) = [\underline{f}_i(x), \bar{f}_i(x)]$ for all $i \in \{1, \dots, m\}$ and let $\leq_\varphi$ be the preference order relation given by (3). Then F is φ -convex if and only if $\Lambda_i^T f$ and $B_i^T f$ are convex for all $i \in \{1, \dots, m\}$, where $f = (\underline{f}_1, \bar{f}_1, \dots, \underline{f}_m, \bar{f}_m)$.

Proof. On the one hand, from definitions of Λ_i , B_i and f , it follows that

$$\Lambda_i^T f = \lambda_{2i-1} \underline{f}_i + \lambda_{2i} \bar{f}_i$$

and

$$B_i^T f = \beta_{2i-1} \underline{f}_i + \beta_{2i} \overline{f}_i.$$

On the other hand, considering (3), the definition of φ -convexity, Proposition 2.3 and considering the definition of the automorphisms $\varphi_i \in \mathcal{A}$, then for all $i \in \{1, \dots, m\}$, it follows that

$$\begin{aligned} F(tx + (1-t)y) &\leq_{\varphi} t \square F(x) \boxplus (1-t) \square F(y) \Leftrightarrow F_i(tx + (1-t)y) \leq_{\varphi_i} t F_i(x) + (1-t) F_i(y) \\ &\Leftrightarrow \tilde{\varphi}_i \left(\left[\underline{f}_i(tx + (1-t)y), \overline{f}_i(tx + (1-t)y) \right] \right) \leq \tilde{\varphi}_i \left(t \left[\underline{f}_i(x), \overline{f}_i(x) \right] + (1-t) \left[\underline{f}_i(y), \overline{f}_i(y) \right] \right) \\ &\Leftrightarrow \tilde{\varphi}_i \left(\left[\underline{f}_i(tx + (1-t)y), \overline{f}_i(tx + (1-t)y) \right] \right) \leq \tilde{\varphi}_i \left(\left[t \underline{f}_i(x) + (1-t) \underline{f}_i(y), t \overline{f}_i(x) + (1-t) \overline{f}_i(y) \right] \right) \\ &\Leftrightarrow \varphi_i \left(\left[\underline{f}_i(tx + (1-t)y), \overline{f}_i(tx + (1-t)y) \right] \right) \leq \varphi_i \left(\left[t \underline{f}_i(x) + (1-t) \underline{f}_i(y), t \overline{f}_i(x) + (1-t) \overline{f}_i(y) \right] \right) \\ &\Leftrightarrow [\lambda_{2i-1} \underline{f}_i + \lambda_{2i} \overline{f}_i](tx + (1-t)y) \leq t[\lambda_{2i-1} \underline{f}_i + \lambda_{2i} \overline{f}_i](x) + (1-t)[\lambda_{2i-1} \underline{f}_i + \lambda_{2i} \overline{f}_i](y) \end{aligned}$$

and

$$[\beta_{2i-1} \underline{f}_i + \beta_{2i} \overline{f}_i](tx + (1-t)y) \leq t[\beta_{2i-1} \underline{f}_i + \beta_{2i} \overline{f}_i](x) + (1-t)[\beta_{2i-1} \underline{f}_i + \beta_{2i} \overline{f}_i](y).$$

Therefore, F is φ -convex if and only if $\Lambda_i^T f$ and $B_i^T f$ are convex for all $i \in \{1, \dots, m\}$. $\square$

Example 3.8 (Scalarization). Associated with (13), consider the weighted scalar problem (see [7])

$$(MOP_{\varphi}(w)) \min_{x \in S} \sum_{i=1}^m (w_{2i-1} \Lambda_i^T f(x) + w_{2i} B_i^T f(x)), \quad (14)$$

where $w = (w_1, \dots, w_{2m}) \in \mathbb{R}^{2m}$, $w_i \geq 0$ for all $i \in \{1, \dots, 2m\}$ and $\sum_{i=1}^{2m} w_i = 1$. The relationships between the solutions for (10) and the solutions for the scalar problem (14) are as follows.

1. If $\bar{x} \in S$ is an optimal solution for $(MOP_{\varphi}(w))$, then $\bar{x} \in S$ is a weak optimal solution for $(1MIOP_{\varphi})$.
2. If $\bar{x} \in S$ is an optimal solution for $(MOP_{\varphi}(w))$ with $w_i > 0$ for all $i \in \{1, \dots, 2m\}$, then $\bar{x} \in S$ is an optimal solution for $(1MIOP_{\varphi})$.
3. If $\bar{x} \in S$ is the only optimal solution for $(MOP_{\varphi}(w))$, then $\bar{x} \in S$ is an optimal solution for $(1MIOP_{\varphi})$.
4. If S is convex, F is φ -convex and $\bar{x} \in S$ is an optimal solution for $(1MIOP_{\varphi})$, then there exists $w = (w_1, \dots, w_{2m}) \in \mathbb{R}^{2m}$ with

$$w_i \geq 0 \text{ for all } i \in \{1, \dots, 2m\} \text{ and } \sum_{i=1}^{2m} w_i = 1 \text{ such that } \bar{x} \in S \text{ is an optimal solution for } (MOP_{\varphi}(w)).$$

Example 3.9 (Necessary and sufficient optimality conditions). Given $F : S \subseteq \mathbb{R}^n \rightarrow (\mathcal{K}_C)^m$ such that $F(x) = (F_1(x), \dots, F_m(x))$ with $F_i(x) = [\underline{f}_i(x), \overline{f}_i(x)]$ for all $i \in \{1, \dots, m\}$, let $\Lambda_i^T f$, $B_i^T f$ be differentiable for all $i \in \{1, \dots, m\}$ and $\mathbf{0} = (0, \dots, 0) \in \mathbb{R}^n$. Thus,

1. If $\bar{x} \in S$ is a weak optimal solution for $(1MIOP_{\varphi})$, then there are $w_1, \dots, w_{2m} \in \mathbb{R}$, with $w_i \geq 0$ not equal zero for all $i \in \{1, \dots, 2m\}$, such that

$$\begin{aligned} \sum_{i=1}^m w_{2i-1} \nabla \Lambda_i^T f(\bar{x}) + \sum_{i=1}^m w_{2i} \nabla B_i^T f(\bar{x}) &= \sum_{i=1}^m \nabla [(w_{2i-1} \Lambda_i^T + w_{2i} B_i^T) f](\bar{x}) \\ &= \nabla \left\{ \sum_{i=1}^m [(w_{2i-1} \Lambda_i^T + w_{2i} B_i^T) f] \right\}(\bar{x}) = \mathbf{0}. \end{aligned} \quad (15)$$

2. If $\bar{x} \in S$, S is convex, F is φ -convex and there are $w_1, \dots, w_{2m} \in \mathbb{R}$, with $w_i \geq 0$ not equal zero for all $i \in \{1, \dots, 2m\}$, such that (15) holds, then $\bar{x}$ is a weak optimal solution for $(1MIOP_{\varphi})$.
3. If $\bar{x} \in S$, S is convex, F is φ -convex and there are $w_1, \dots, w_{2m} \in \mathbb{R}$, with $w_i > 0$ for all $i \in \{1, \dots, 2m\}$, such that (15) holds, then $\bar{x}$ is an optimal solution for $(1MIOP_{\varphi})$.

We note that if $\underline{f}_i$ and $\overline{f}_i$ are differentiable for all $i \in \{1, \dots, m\}$, then (15) can be rewritten as being

$$\sum_{i=1}^m (w_{2i-1} \lambda_{2i-1} \nabla \underline{f}_i(\bar{x}) + w_{2i-1} \lambda_{2i} \nabla \overline{f}_i(\bar{x})) + \sum_{i=1}^m (w_{2i} \beta_{2i-1} \nabla \underline{f}_i(\bar{x}) + w_{2i} \beta_{2i} \nabla \overline{f}_i(\bar{x}))$$

$$= \sum_{i=1}^m \left((w_{2i-1} \lambda_{2i-1} + w_{2i-1} \beta_{2i-1}) \nabla \underline{f}_i(\bar{x}) \right) + \sum_{i=1}^m \left((w_{2i-1} \lambda_{2i} + w_{2i-1} \beta_{2i}) \nabla \bar{f}_i(\bar{x}) \right) = 0. \quad (16)$$

However, due to the algebraic structure on $\mathcal{K}_C$, in many cases it is very restrictive to require $\underline{f}_i$ and $\bar{f}_i$ to be differentiable for all $i \in \{1, \dots, m\}$. In fact, consider the interval-valued function $F : \mathbb{R} \rightarrow (\mathcal{K}_C)^2$, defined by $F(x) = (F_1(x), F_2(x))$, where $F_1(x) = [-|x|, |x|] = [\underline{f}_1(x), \bar{f}_1(x)]$ and $F_2(x) = [0, x^2] = [\underline{f}_2(x), \bar{f}_2(x)]$. Hence, $\underline{f}_1$ and $\bar{f}_1$ are not differentiable at $x = 0$, although $x = 0$ is a strict minimum for F considering the order relation $\leq_\varphi$ with φ given in Example 2.2. On the other hand, we have that $\sum_{i=1}^2 (\Lambda_i^T f + B_i^T f) = (\Lambda_i^T + B_i^T) f : \mathbb{R} \rightarrow \mathbb{R}$ is given by $\sum_{i=1}^2 [(\Lambda_i^T + B_i^T) f](x) = x^2$ for all $x \in \mathbb{R}$ since, for this case, $\Lambda_1 = (1, 0, 0, 0)$, $\Lambda_2 = (0, 0, 1, 0)$, $B_1 = (0, 1, 0, 0)$, $B_2 = (0, 0, 0, 1)$ and $f(x) = (-|x|, |x|, 0, x^2)$. Therefore, $\sum_{i=1}^2 (\Lambda_i^T f + B_i^T f)$ is a differentiable and convex function and $\nabla \left(\sum_{i=1}^2 (\Lambda_i^T + B_i^T) f \right)(0) = 0$, which is a version of condition (15).

4. Fuzzy spaces: preliminaries

A fuzzy subset A of $\mathbb{R}^m$, given an positive integer m , is characterized by a function $\tilde{u} : \mathbb{R}^m \rightarrow [0, 1]$ called the membership function of A . In general, in order to simplify the notation, a fuzzy subset A of $\mathbb{R}^m$ is presented as being its membership function $\tilde{u}$. That is, a fuzzy subset A of $\mathbb{R}$ is a function $\tilde{u} : \mathbb{R} \rightarrow [0, 1]$. In this study this representation is adopted. Moreover, the family of all fuzzy subsets of $\mathbb{R}^m$ is denoted by $\mathcal{F}(\mathbb{R}^m)$.

The level sets of a given $\tilde{u} \in \mathcal{F}(\mathbb{R}^m)$, are defined by $[\tilde{u}]^\alpha = \{x \in \mathbb{R}^m : \alpha \leq \tilde{u}(x)\}$ for all $\alpha \in (0, 1]$ and by $[\tilde{u}]^0 = \text{cl}\{x \in \mathbb{R}^m : 0 < \tilde{u}(x)\}$, where $\text{cl}(A)$ denotes the closure of a classical set $A \subset \mathbb{R}^m$. These sets are called the α -level sets of $\tilde{u}$ for all $\alpha \in [0, 1]$.

The family of all fuzzy subset $\tilde{u}$ of $\mathbb{R}^m$ that has the following properties is denoted by $\mathcal{F}_C(\mathbb{R}^m)$:

- (i) $\tilde{u}$ is normal, i.e., there exists $x^* \in \mathbb{R}$ such that $\tilde{u}(x^*) = 1$;
- (ii) $\tilde{u}$ is fuzzy convex, i.e., $\min\{\tilde{u}(x), \tilde{u}(y)\} \leq \tilde{u}(tx + (1-t)y)$ for all $x, y \in \mathbb{R}^m$ and for all $t \in [0, 1]$;
- (iii) $\tilde{u}$ is upper semicontinuous on $\mathbb{R}^m$, i.e., given $x^* \in \mathbb{R}^m$, for every $\epsilon > 0$ there exists $\delta > 0$ such that $\tilde{u}(x) - \tilde{u}(x^*) < \epsilon$ for all $x \in \mathbb{R}^m$ with $|x - x^*| < \delta$;
- (iv) $\tilde{u}$ is compactly supported, i.e., $\text{cl}\{x \in \mathbb{R}^m : 0 < \tilde{u}(x)\}$ is compact.

If $m = 1$, then $\tilde{u} \in \mathcal{F}_C(\mathbb{R})$ is said to be a real fuzzy interval and $[\tilde{u}]^\alpha \in \mathcal{K}_C$ for all $\alpha \in [0, 1]$ (see. e.g., [10,12]).

This study focuses on the subfamily $(\mathcal{F}_C(\mathbb{R}))^m \subset \mathcal{F}_C(\mathbb{R}^m)$, where $(\mathcal{F}_C(\mathbb{R}))^m$ denotes the Cartesian product composed of m -terms $\mathcal{F}_C(\mathbb{R})$, i.e.,

$$(\mathcal{F}_C(\mathbb{R}))^m = \mathcal{F}_C(\mathbb{R}) \times \mathcal{F}_C(\mathbb{R}) \times \dots \times \mathcal{F}_C(\mathbb{R}).$$

Given $\tilde{u} = (\tilde{u}_1, \dots, \tilde{u}_m) \in (\mathcal{F}_C(\mathbb{R}))^m$, it follows that [13]

$$[\tilde{u}]^\alpha = \{(x_1, \dots, x_m) \in \mathbb{R}^m : \tilde{u}_i(x_i) \geq \alpha, i \in \{1, \dots, m\}\}.$$

Geometrically, it would be a Cartesian product of closed intervals in $\mathbb{R}^m$

$$[\tilde{u}]^\alpha = \prod_{i=1}^m [\tilde{u}_i]^\alpha. \quad (17)$$

It is well-known that, given $\lambda \in \mathbb{R}$ and $\tilde{a}, \tilde{b} \in \mathcal{F}_C(\mathbb{R})$, then the addition $\tilde{a} \oplus \tilde{b}$ and the multiplication by scalar $\lambda \odot \tilde{a}$ can be characterized level-wise, respectively, by

$$[\tilde{a} \oplus \tilde{b}]^\alpha = [\tilde{a}]^\alpha + [\tilde{b}]^\alpha, \text{ and } [\lambda \odot \tilde{a}]^\alpha = \lambda \cdot [\tilde{a}]^\alpha$$

for all $\alpha \in [0, 1]$. Thus, by considering the algebraic operations “ $\tilde{\boxplus}$ ” and “ $\tilde{\boxtimes}$ ” on $(\mathcal{F}_C(\mathbb{R}))^m$ given by

$$(\tilde{u}_1, \dots, \tilde{u}_m) \tilde{\boxplus} (\tilde{v}_1, \dots, \tilde{v}_m) = (\tilde{u}_1 \oplus \tilde{v}_1, \dots, \tilde{u}_m \oplus \tilde{v}_m)$$

and

$$\lambda \tilde{\boxtimes} (\tilde{u}_1, \dots, \tilde{u}_m) = (\lambda \odot \tilde{u}_1, \dots, \lambda \odot \tilde{u}_m),$$

respectively, for all $\tilde{u} = (\tilde{u}_1, \dots, \tilde{u}_m), \tilde{v} = (\tilde{v}_1, \dots, \tilde{v}_m) \in (\mathcal{F}_C(\mathbb{R}))^m$ and for all $\lambda \in \mathbb{R}$, then from (17), it follows that these operations can be characterized level-wise, respectively, by

$$[\tilde{u} \tilde{\boxplus} \tilde{v}]^\alpha = \prod_{i=1}^m [\tilde{u}_i \oplus \tilde{v}_i]^\alpha$$

and

$$[\lambda \tilde{\square} \tilde{a}]^\alpha = \prod_{i=1}^m [\lambda \odot \tilde{u}_i]^\alpha$$

for all $\alpha \in [0, 1]$. Another well-known fact is that, $(\mathcal{F}_C(\mathbb{R}), D)$ is a complete metric space (see, e.g., [10]), where $D : \mathcal{F}_C(\mathbb{R}) \times \mathcal{F}_C(\mathbb{R}) \rightarrow \mathbb{R}$ is given by

$$D(\tilde{a}, \tilde{b}) = \sup_{\alpha \in [0, 1]} \max\{H([\tilde{a}]^\alpha, [\tilde{b}]^\alpha)\}.$$

Thus, $((\mathcal{F}_C(\mathbb{R}))^m, D_H)$ is also a complete metric space, where $D_H : (\mathcal{F}_C(\mathbb{R}))^m \times (\mathcal{F}_C(\mathbb{R}))^m \rightarrow \mathbb{R}$ is given by

$$D_H((\tilde{a}_1, \dots, \tilde{a}_m), (\tilde{b}_1, \dots, \tilde{b}_m)) = \max\{D(\tilde{a}_1, \tilde{b}_1), \dots, D(\tilde{a}_m, \tilde{b}_m)\}.$$

There are several concepts and results in fuzzy set theory that are presented through the concept of α -level sets of fuzzy sets, as those presented above. Using this approach the concept of $\varphi, \leq_{\mathbb{R}^{2m}}$ -preference order relation on $(\mathcal{F}_C(\mathbb{R}))^m$ is introduced as follows.

Definition 4.1. The $\varphi, \leq_{\mathbb{R}^{2m}}$ -preference order relation $\tilde{\leq}_\varphi^{2m}$ on $(\mathcal{F}_C(\mathbb{R}))^m$ is given level-wise by

$$\begin{aligned} (\tilde{u}_1, \dots, \tilde{u}_m) \tilde{\leq}_\varphi^{2m} (\tilde{v}_1, \dots, \tilde{v}_m) &\Leftrightarrow ([\tilde{u}_1]^\alpha, \dots, [\tilde{u}_m]^\alpha) \leq_\varphi^{2m} ([\tilde{v}_1]^\alpha, \dots, [\tilde{v}_m]^\alpha) \\ &\Leftrightarrow \tilde{\varphi}([\tilde{u}_1]^\alpha, \dots, [\tilde{u}_m]^\alpha) \leq_{\mathbb{R}^{2m}} \tilde{\varphi}([\tilde{v}_1]^\alpha, \dots, [\tilde{v}_m]^\alpha) \end{aligned}$$

for all $\alpha \in [0, 1]$, where $(\tilde{u}_1, \dots, \tilde{u}_m), (\tilde{v}_1, \dots, \tilde{v}_m) \in (\mathcal{F}_C(\mathbb{R}))^m$ and $\leq_\varphi^{2m}$ is the order relation given in (1).

The following family of preference order relations, dependent on φ , considering $\varphi \in \mathcal{A}^m$ and $\leq_{\mathbb{R}^{2m}}$ the order relation on $\mathbb{R}^{2m}$ given by (2), is defined by

$$(\tilde{u}_1, \dots, \tilde{u}_m) \tilde{\leq}_\varphi (\tilde{v}_1, \dots, \tilde{v}_m) \Leftrightarrow ([\tilde{u}_1]^\alpha, \dots, [\tilde{u}_m]^\alpha) \leq_\varphi ([\tilde{v}_1]^\alpha, \dots, [\tilde{v}_m]^\alpha) \quad (18)$$

for all $\alpha \in [0, 1]$, where $(\tilde{u}_1, \dots, \tilde{u}_m), (\tilde{v}_1, \dots, \tilde{v}_m) \in (\mathcal{F}_C(\mathbb{R}))^m$ and $\leq_\varphi$ is the order relation given in (3).

Proposition 4.1. The preference order relation $\tilde{\leq}_\varphi$ given by (18) is a partial order relation on $(\mathcal{F}_C(\mathbb{R}))^m$.

Proof. This result follows directly from Proposition 2.2 and from (18). $\square$

The strict order relations $\tilde{<}_\varphi$ and $\tilde{<}_\varphi$, considering the partial order relation (18), are defined on $(\mathcal{F}_C(\mathbb{R}))^m$ as follows:

$$\begin{aligned} (\tilde{u}_1, \dots, \tilde{u}_m) \tilde{<}_\varphi (\tilde{v}_1, \dots, \tilde{v}_m) &\Leftrightarrow \begin{cases} (\tilde{u}_1, \dots, \tilde{u}_m) \tilde{\leq}_\varphi (\tilde{v}_1, \dots, \tilde{v}_m) \\ \text{and} \\ (\tilde{u}_1, \dots, \tilde{u}_m) \neq (\tilde{v}_1, \dots, \tilde{v}_m) \end{cases} \\ &\Leftrightarrow \begin{cases} ([\tilde{u}_1]^\alpha, \dots, [\tilde{u}_m]^\alpha) \leq_\varphi ([\tilde{v}_1]^\alpha, \dots, [\tilde{v}_m]^\alpha) \\ \text{and} \\ ([\tilde{u}_1]^\alpha, \dots, [\tilde{u}_m]^\alpha) \neq ([\tilde{v}_1]^\alpha, \dots, [\tilde{v}_m]^\alpha) \end{cases} \\ &\Leftrightarrow \begin{cases} [\tilde{u}_i]^\alpha \leq_{\varphi_i} [\tilde{v}_i]^\alpha \quad \forall \alpha \in [0, 1] \text{ and } \forall i \in \{1, \dots, m\} \\ \text{and} \\ \exists \gamma \in [0, 1] \text{ and } \exists k \in \{1, \dots, m\} \text{ such that } [\tilde{u}_k]^\gamma \neq [\tilde{v}_k]^\gamma \end{cases}, \end{aligned} \quad (19)$$

and

$$\begin{aligned} (\tilde{u}_1, \dots, \tilde{u}_m) \tilde{<}_\varphi (\tilde{v}_1, \dots, \tilde{v}_m) &\Leftrightarrow ([\tilde{u}_1]^\alpha, \dots, [\tilde{u}_m]^\alpha) <_\varphi ([\tilde{v}_1]^\alpha, \dots, [\tilde{v}_m]^\alpha) \\ &\Leftrightarrow [\tilde{u}_i]^\alpha <_{\varphi_i} [\tilde{v}_i]^\alpha, \end{aligned} \quad (20)$$

for all $i \in \{1, \dots, m\}$ and for all $\alpha \in [0, 1]$.

The next example illustrates the applicability of this class of preference order relations. In order to understand completely the example the reader need to have familiarity with the concept of level sets of triangle and trapezoidal fuzzy intervals, which can be found in [3].

Example 4.1. Failure mode and effects analysis (FMEA) is a methodology that allows the identification and study of the cause and effects of possible failures in a production or in project phases of a product. The traditional FMEA determines the risk priorities

Table 1
From Table 5 in [29].

Fuzzy ratings for occurrence of a failure		
Rating	Probability of occurrence	Fuzzy interval
Very high (VH)	Failure is almost inevitable	(8; 9; 10; 10)
High (H)	Repeated failures	(6; 7; 8; 9)
Moderate (M)	Occasional failures	(3; 4; 6; 7)
Low (L)	Relatively few failures	(1; 2; 3; 4)
Remote (R)	Failure is unlikely	(1; 1; 2)

Table 2
From Table 6 in [29].

Fuzzy ratings for severity of a failure		
Rating	Severity of effect	Fuzzy interval
Hazardous without warning (HWOW)	Very high severity ranking without warning	(9; 10; 10)
Hazardous with warning (HWW)	Very high severity ranking with warning	(8; 9; 10)
Very high (VH)	System inoperable with destructive failure	(7; 8; 9)
High (H)	System inoperable with equipment damage	(6; 7; 8)
Moderate (M)	System inoperable with minor damage	(5; 6; 7)
Low (L)	System inoperable without damage	(4; 5; 6)
Very low (VL)	System operable with significant degradation of performance	(3; 4; 5)
Minor (MR)	System operable with some degradation of performance	(2; 3; 4)
Very minor (VMR)	System operable with minimal interference	(1; 2; 3)
None (N)	No effect	(1; 1; 2)

Table 3
From Table 7 in [29].

Fuzzy ratings for detection of a failure		
Rating	Likelihood of detection	Fuzzy interval
Absolute uncertainty (AU)	No chance	(9; 10; 10)
Very remote (VR)	Very remote chance	(8; 9; 10)
Remote (R)	Remote chance	(7; 8; 9)
Very Low (VL)	Very low chance	(6; 7; 8)
Low (L)	Low chance	(5; 6; 7)
Moderate (M)	Moderate chance	(4; 5; 6)
Moderately high (MH)	Moderately high chance	(3; 4; 5)
high (H)	High chance	(2; 3; 4)
Very high (VH)	Very high chance	(1; 2; 3)
Almost certain (AC)	Almost certainty	(1; 1; 2)

of failure modes based on the risk factors like the occurrence (O), severity (S) and detection (D) of each failure mode. However, risk factors O, S and D are not easy to accurately assess and, for this reason, the idea of analyzing them in a linguistic way has been adopted in some studies, such as [29]. Each one of these linguistic variables can be modeled by means of a fuzzy interval as illustrated in Tables 1–3, which were introduced as Tables 5–7 in [29].

Considering $(\tilde{u}_1, \tilde{u}_2, \tilde{u}_3), (\tilde{v}_1, \tilde{v}_2, \tilde{v}_3) \in (\mathcal{F}_C(\mathbb{R}))^3$ such that $\tilde{u}_1$ and $\tilde{v}_1$ represent fuzzy ratings for occurrence of a failure, $\tilde{u}_2$ and $\tilde{v}_2$ represent fuzzy ratings for severity of a failure and $\tilde{u}_3$ and $\tilde{v}_3$ represent fuzzy ratings for detection of a failure, these 3-tuples of fuzzy intervals allow the representation of different scenarios, such as

$$(\tilde{u}_1, \tilde{u}_2, \tilde{u}_3) = (\text{Very high, Hazardous with warning, high}) = ((8; 9; 10; 10), (8; 9; 10), (2; 3; 4))$$

and

$$(\tilde{v}_1, \tilde{v}_2, \tilde{v}_3) = (\text{Moderate, Hazardous without warning, Low}) = ((3; 4; 6; 7), (9; 10; 10), (5; 6; 7)).$$

A decision for the preferable one needs to be done, whenever it is possible to compare both. In this sense, the preference order relation $\tilde{\preceq}_\varphi$ helps to provide a criterion for such comparison. Indeed, given $\varphi = (\varphi_1 \times \varphi_2 \times \varphi_3) \in \mathcal{A}^3$, if $(\tilde{u}_1, \tilde{u}_2, \tilde{u}_3)$ and $(\tilde{v}_1, \tilde{v}_2, \tilde{v}_3)$ are comparable via $\tilde{\preceq}_\varphi$, then one can say that the scenarios $(\tilde{u}_1, \tilde{u}_2, \tilde{u}_3)$ are preferable to $(\tilde{v}_1, \tilde{v}_2, \tilde{v}_3)$ whenever

$$(\tilde{u}_1, \tilde{u}_2, \tilde{u}_3) \tilde{\preceq}_\varphi (\tilde{v}_1, \tilde{v}_2, \tilde{v}_3).$$

Obviously, this comparison depends on the choice of $\varphi = (\varphi_1 \times \varphi_2 \times \varphi_3) \in \mathcal{A}^3$, which can be seen as decision making. Supposing that such decision making is based on

$$\varphi_1(x_1, x_2) = \left(\frac{x_2 - x_1}{2}, \frac{x_1 + x_2}{2}\right), \text{ where } \lambda_1 = -\frac{1}{2}, \lambda_2 = \frac{1}{2}, \beta_1 = \frac{1}{2} = \beta_2;$$

$$\varphi_2(x_3, x_4) = (x_3, x_4), \text{ where } \lambda_3 = 1, \lambda_4 = 0, \beta_3 = 0, \beta_4 = 1;$$

$$\varphi_3(x_5, x_6) = (x_5, x_6 - x_5), \text{ where } \lambda_5 = 1, \lambda_6 = 0, \beta_5 = -1, \beta_6 = 1,$$

it follows that $(\tilde{u}_1, \tilde{u}_2, \tilde{u}_3)$ is preferable to $(\tilde{v}_1, \tilde{v}_2, \tilde{v}_3)$, since

$$\bar{\varphi}_1([\tilde{u}_1]^\alpha) = \bar{\varphi}_1([(8; 9; 10; 10)]^\alpha) = \bar{\varphi}_1([8 + \alpha, 10]) = \left(\frac{2 - \alpha}{2}, \frac{18 + \alpha}{2}\right) \forall \alpha \in [0, 1],$$

$$\bar{\varphi}_1([\tilde{v}_1]^\alpha) = \bar{\varphi}_1([(3; 4; 6; 7)]^\alpha) = \bar{\varphi}_1([3 + \alpha, 7 - \alpha]) = \left(\frac{4 - 2\alpha}{2}, 10\right) \forall \alpha \in [0, 1],$$

$$\bar{\varphi}_2([\tilde{u}_2]^\alpha) = \bar{\varphi}_2([(8; 9; 10)]^\alpha) = \bar{\varphi}_2([8 + \alpha, 10 - \alpha]) = (8 + \alpha, 10 - \alpha) \forall \alpha \in [0, 1],$$

$$\bar{\varphi}_2([\tilde{v}_2]^\alpha) = \bar{\varphi}_2([(9; 10; 10)]^\alpha) = \bar{\varphi}_2([9 + \alpha, 10]) = (9 + \alpha, 10) \forall \alpha \in [0, 1],$$

$$\bar{\varphi}_3([\tilde{u}_3]^\alpha) = \bar{\varphi}_3([(2; 3; 4)]^\alpha) = \bar{\varphi}_3([2 + \alpha, 4 - \alpha]) = (2 + \alpha, 2 - 2\alpha) \forall \alpha \in [0, 1],$$

and

$$\bar{\varphi}_3([\tilde{v}_3]^\alpha) = \bar{\varphi}_3([(5; 6; 7)]^\alpha) = \bar{\varphi}_3([5 + \alpha, 7 - \alpha]) = (5 + \alpha, 2 - 2\alpha) \forall \alpha \in [0, 1],$$

from where $\bar{\varphi}_i([\tilde{u}_i]^\alpha) \leq \bar{\varphi}_i([\tilde{v}_i]^\alpha)$, i.e., $[\tilde{u}_i]^\alpha \leq_{\varphi_i} [\tilde{v}_i]^\alpha$ for all $\alpha \in [0, 1]$ and for all $i \in \{1, 2, 3\}$. Consequently,

$$([\tilde{u}_1]^\alpha, [\tilde{u}_2]^\alpha, [\tilde{u}_3]^\alpha) \leq_\varphi ([\tilde{v}_1]^\alpha, [\tilde{v}_2]^\alpha, [\tilde{v}_3]^\alpha)$$

for all $\alpha \in [0, 1]$. Therefore, $(\tilde{u}_1, \tilde{u}_2, \tilde{u}_3) \leq_\varphi (\tilde{v}_1, \tilde{v}_2, \tilde{v}_3)$.

Example 4.2. Considering $\varphi \in \mathcal{A}^m$ given in Example 2.2, then for all $\alpha \in [0, 1]$ and for all $i \in \{1, \dots, m\}$, it follows that

$$(\tilde{u}_1, \dots, \tilde{u}_m) \leq_\varphi (\tilde{v}_1, \dots, \tilde{v}_m) \Leftrightarrow ([\tilde{u}_1]^\alpha, \dots, [\tilde{u}_m]^\alpha) \leq_\varphi ([\tilde{v}_1]^\alpha, \dots, [\tilde{v}_m]^\alpha)$$

$$\Leftrightarrow [\tilde{u}_i]^\alpha \leq_{\varphi_i} [\tilde{v}_i]^\alpha$$

$$\Leftrightarrow (\underline{u}_{i,\alpha}, \bar{u}_{i,\alpha}) \leq (\underline{v}_{i,\alpha}, \bar{v}_{i,\alpha})$$

$$\Leftrightarrow \underline{u}_{i,\alpha} \leq \underline{v}_{i,\alpha} \text{ and } \bar{u}_{i,\alpha} \leq \bar{v}_{i,\alpha},$$

where $[\tilde{u}_i]^\alpha = [\underline{u}_{i,\alpha}, \bar{u}_{i,\alpha}]$ and $[\tilde{v}_i]^\alpha = [\underline{v}_{i,\alpha}, \bar{v}_{i,\alpha}]$ for all $\alpha \in [0, 1]$ and $i \in \{1, \dots, m\}$.

Example 4.3. Considering $\varphi \in \mathcal{A}^m$ given in Example 2.4, then for all $\alpha \in [0, 1]$ and for all $i \in \{1, \dots, m\}$ it follows that

$$(\tilde{u}_1, \dots, \tilde{u}_m) \leq_\varphi (\tilde{v}_1, \dots, \tilde{v}_m) \Leftrightarrow ([\tilde{u}_1]^\alpha, \dots, [\tilde{u}_m]^\alpha) \leq_\varphi ([\tilde{v}_1]^\alpha, \dots, [\tilde{v}_m]^\alpha)$$

$$\Leftrightarrow [\tilde{u}_i]^\alpha \leq_{\varphi_i} [\tilde{v}_i]^\alpha$$

$$\Leftrightarrow \left(\frac{\underline{u}_{i,\alpha} + \bar{u}_{i,\alpha}}{2}, \frac{\bar{u}_{i,\alpha} - \underline{u}_{i,\alpha}}{2}\right) \leq \left(\frac{\underline{v}_{i,\alpha} + \bar{v}_{i,\alpha}}{2}, \frac{\bar{v}_{i,\alpha} - \underline{v}_{i,\alpha}}{2}\right)$$

$$\Leftrightarrow \frac{\underline{u}_{i,\alpha} + \bar{u}_{i,\alpha}}{2} \leq \frac{\underline{v}_{i,\alpha} + \bar{v}_{i,\alpha}}{2} \text{ and } \frac{\bar{u}_{i,\alpha} - \underline{u}_{i,\alpha}}{2} \leq \frac{\bar{v}_{i,\alpha} - \underline{v}_{i,\alpha}}{2},$$

where $[\tilde{u}_i]^\alpha = [\underline{u}_{i,\alpha}, \bar{u}_{i,\alpha}]$ and $[\tilde{v}_i]^\alpha = [\underline{v}_{i,\alpha}, \bar{v}_{i,\alpha}]$ for all $\alpha \in [0, 1]$ and for all $i \in \{1, \dots, m\}$.

Example 4.4. Considering $\varphi \in \mathcal{A}^m$ given in Example 2.3, then for all $\alpha \in [0, 1]$ and for all $i \in \{1, \dots, m\}$ it follows that

$$(\tilde{u}_1, \dots, \tilde{u}_m) \leq_\varphi (\tilde{v}_1, \dots, \tilde{v}_m) \Leftrightarrow ([\tilde{u}_1]^\alpha, \dots, [\tilde{u}_m]^\alpha) \leq_\varphi ([\tilde{v}_1]^\alpha, \dots, [\tilde{v}_m]^\alpha)$$

$$\Leftrightarrow [\tilde{u}_i]^\alpha \leq_{\varphi_i} [\tilde{v}_i]^\alpha$$

$$\Leftrightarrow (\underline{u}_{i,\alpha}, \bar{u}_{i,\alpha} - \underline{u}_{i,\alpha}) \leq (\underline{v}_{i,\alpha}, \bar{v}_{i,\alpha} - \underline{v}_{i,\alpha})$$

$$\Leftrightarrow \underline{u}_{i,\alpha} \leq \underline{v}_{i,\alpha} \text{ and } \bar{u}_{i,\alpha} - \underline{u}_{i,\alpha} \leq \bar{v}_{i,\alpha} - \underline{v}_{i,\alpha},$$

where $[\tilde{u}_i]^\alpha = [\underline{u}_{i,\alpha}, \bar{u}_{i,\alpha}]$ and $[\tilde{v}_i]^\alpha = [\underline{v}_{i,\alpha}, \bar{v}_{i,\alpha}]$ for all $\alpha \in [0, 1]$ and for all $i \in \{1, \dots, m\}$.

The preference order relations given in (18), (19) and (20) are considered in the formulation of the φ -multiobjective fuzzy interval optimization problems and its concepts of solution presented in Section 5. Also are considered the maps $\tilde{F} : S \subseteq \mathbb{R}^n \rightarrow (\mathcal{F}_C(\mathbb{R}))^m$ given by $\tilde{F}(x) = (\tilde{F}_1(x), \dots, \tilde{F}_m(x))$, with $[\tilde{F}_i(x)]^\alpha(x) = [\underline{f}_{i,\alpha}(x), \bar{f}_{i,\alpha}(x)]$, where n and m are positive integers and $\underline{f}_{i,\alpha}, \bar{f}_{i,\alpha} : S \rightarrow \mathbb{R}$ are multivariable real-valued functions such that $\underline{f}_{i,\alpha}(x) \leq \bar{f}_{i,\alpha}(x)$ for all $x \in S$, for all $i \in \{1, \dots, m\}$ and for all $\alpha \in [0, 1]$. $\tilde{F}$ is called

an *multi-fuzzy interval-valued function* whenever $m > 1$. If $m = 1$, then $\tilde{F}$ is said to be a fuzzy interval-valued function. The functions $\tilde{F}_1, \dots, \tilde{F}_m$ are called the *fuzzy interval component functions of $\tilde{F}$* .

Remark 4.1. From (17), it is easy to see that given a multi-fuzzy interval-valued function $F : S \subseteq \mathbb{R}^n \rightarrow (\mathcal{F}_C(\mathbb{R}))^m$ for $\tilde{F}(x) = (\tilde{F}_1(x), \dots, \tilde{F}_m(x))$, with $[\tilde{F}_i(x)]^\alpha(x) = [\underline{f}_{i,\alpha}(x), \overline{f}_{i,\alpha}(x)]$ for all $\alpha \in [0, 1]$, for all $x \in S$ and for all $i \in \{1, \dots, m\}$, it follows that

$$\begin{aligned} [\tilde{F}(x)]^\alpha &= ([\tilde{F}_1(x), \dots, \tilde{F}_m(x)])^\alpha = ([\tilde{F}_1(x)]^\alpha, \dots, [\tilde{F}_m(x)]^\alpha) \\ &= ([\underline{f}_{1,\alpha}(x), \overline{f}_{1,\alpha}(x)], \dots, [\underline{f}_{m,\alpha}(x), \overline{f}_{m,\alpha}(x)]) \in (\mathcal{K}_C)^m. \end{aligned}$$

The concepts of limit and continuity for multi-fuzzy interval-valued functions are naturally defined on the metric space $((\mathcal{F}_C(\mathbb{R}))^m, D_H)$. In addition, the next concepts and results are used to provide, in a concise way, some optimality conditions for the φ -multiobjective fuzzy interval optimization problems that are introduced in Section 5.

Definition 4.2. Let $\tilde{\leq}_\varphi$ be the preference order relation given by (18) and let $\tilde{F} : S \subseteq \mathbb{R}^n \rightarrow (\mathcal{F}_C(\mathbb{R}))^m$ be a multi-fuzzy interval-valued function. $\tilde{F}$ is said to be φ -convex if

$$\tilde{F}(tx + (1-t)y) \tilde{\leq}_\varphi t \tilde{\boxplus} \tilde{F}(x) \tilde{\boxplus} (1-t) \tilde{\boxplus} \tilde{F}(y) \quad (21)$$

for all $x, y \in S$ and $t \in (0, 1)$. If there exists $x^* \in S$ such that

$$\tilde{F}(tx^* + (1-t)y) \tilde{\leq}_\varphi t \tilde{\boxplus} \tilde{F}(x^*) \tilde{\boxplus} (1-t) \tilde{\boxplus} \tilde{F}(y) \quad (22)$$

for all $y \in S$ and $t \in (0, 1)$, then $\tilde{F}$ is said to be φ -convex at x^*

The next results follow from (18), definitions of “ $\tilde{\boxplus}$ ” and “ $\tilde{\boxminus}$ ”, from Proposition 2.3, and Corollary 2.1.

Proposition 4.2. Let $\tilde{\leq}_\varphi$ be the preference order relation given by (18) and let $F : S \subseteq \mathbb{R}^n \rightarrow (\mathcal{F}_C(\mathbb{R}))^m$ be a multi-fuzzy interval-valued function such that $\tilde{F}(x) = (\tilde{F}_1(x), \dots, \tilde{F}_m(x))$. $\tilde{F}$ is φ -convex if and only if $\tilde{F}_i$ is φ_i -convex for all $i \in \{1, \dots, m\}$.

Corollary 4.1. Let $\tilde{\leq}_\varphi$ be the preference order relation given by (18) and let $\tilde{F} : S \subseteq \mathbb{R}^n \rightarrow (\mathcal{F}_C(\mathbb{R}))^m$ be a multi-fuzzy interval-valued function. Then $\tilde{F}$ is φ -convex if and only if $[\tilde{F}]^\alpha : S \rightarrow (\mathcal{K}_C)^m$, given by $[F]^\alpha(x) = [F(x)]^\alpha$, it is φ -convex for all $\alpha \in [0, 1]$.

Proof. From definitions of $\tilde{\leq}_\varphi$, “ $\tilde{\boxplus}$ ” and “ $\tilde{\boxminus}$ ”, and from (17), it follows that

$$\begin{aligned} &\tilde{F}(tx + (1-t)y) \tilde{\leq}_\varphi t \tilde{\boxplus} \tilde{F}(x) \tilde{\boxplus} (1-t) \tilde{\boxplus} \tilde{F}(y) \\ \Leftrightarrow &[\tilde{F}]^\alpha(tx + (1-t)y) \leq_\varphi t \boxplus [\tilde{F}]^\alpha(x) \boxplus (1-t) \boxplus [\tilde{F}]^\alpha(y) \quad \forall \alpha \in [0, 1]. \end{aligned}$$

Therefore, $\tilde{F}$ is φ -convex if and only if $[\tilde{F}]^\alpha$ for all $\alpha \in [0, 1]$. $\square$

Corollary 4.2. Given $x^* \in S \subseteq \mathbb{R}^n$, let $\tilde{\leq}_\varphi$ be the preference order relation given by (18) and let $F : S \subseteq \mathbb{R}^n \rightarrow (\mathcal{F}_C(\mathbb{R}))^m$ be a multi-fuzzy interval-valued function such that $\tilde{F}(x) = (\tilde{F}_1(x), \dots, \tilde{F}_m(x))$. $\tilde{F}$ is φ -convex at x^* if and only if $\tilde{F}_i$ is φ_i -convex at x^* for all $i \in \{1, \dots, m\}$.

Remark 4.2. Given a multi-fuzzy interval-valued function $\tilde{F} : S \subseteq \mathbb{R}^n \rightarrow (\mathcal{F}_C(\mathbb{R}))^m$ such that $\tilde{F}(x) = (\tilde{F}_1(x), \dots, \tilde{F}_m(x))$, with $[\tilde{F}_i(x)]^\alpha(x) = [\underline{f}_{i,\alpha}(x), \overline{f}_{i,\alpha}(x)]$ for all $x \in S$, for all $i \in \{1, \dots, m\}$, and for all $\alpha \in [0, 1]$, then from Proposition 4.2 and from definition of the preference order given by (18), it follows that $\tilde{F}$ is φ -convex if and only if the real-valued functions

$$[\lambda_{2i-1} \underline{f}_{i,\alpha} + \lambda_{2i} \overline{f}_{i,\alpha}], [\beta_{2i-1} \underline{f}_{i,\alpha} + \beta_{2i} \overline{f}_{i,\alpha}] : S \rightarrow \mathbb{R}$$

are convex for all $i \in \{1, \dots, m\}$ and for all $\alpha \in [0, 1]$. In particular, $\tilde{F}$ is φ -convex at $x^* \in S$ if and only if the real-valued functions $[\lambda_{2i-1} \underline{f}_{i,\alpha} + \lambda_{2i} \overline{f}_{i,\alpha}]$ and $[\beta_{2i-1} \underline{f}_{i,\alpha} + \beta_{2i} \overline{f}_{i,\alpha}]$ are convex at x^* for all $i \in \{1, \dots, m\}$ and for all $\alpha \in [0, 1]$.

5. Multiobjective fuzzy interval optimization problems

There are many phenomena involving uncertainties that are not easy to accurately quantify. In some of these cases the use of linguistic variables, such as illustrated in Example 4.1, helps in the study of these phenomena by means of mathematical models based on fuzzy theory. The multiobjective fuzzy optimization problem is one important example of these mathematical models that has applicability in several applied problems as in water resources management [4], in power plant maintenance scheduling [6], in production planning problems [11], among others, which is one of the reasons why it has been studied extensively over the last few decades (see, e.g., [13,19,32,24]).

We consider in this study, the following multiobjective fuzzy interval minimization problem denoted by $(1MFIO\varphi)$ and called φ -multiobjective fuzzy interval minimization problem

$$(1MFIO\varphi) \min_{x \in S} \tilde{F}(x), \quad (23)$$

where $\varphi \in \mathcal{A}^m$ and $\tilde{F} : S \subseteq \mathbb{R}^n \rightarrow (\mathcal{F}_C(\mathbb{R}))^m$ is a multi-fuzzy-interval-valued function given by $\tilde{F}(x) = (\tilde{F}_1(x), \dots, \tilde{F}_m(x))$, with $[\tilde{F}_i(x)]^\alpha(x) = [\underline{f}_{i,\alpha}(x), \overline{f}_{i,\alpha}(x)]$ for all $x \in S$, for all $\alpha \in [0, 1]$, and for all $i \in \{1, \dots, m\}$. Based on the preference ordering relation (18), both concepts of local and global solutions for (23) are defined as follows.

Definition 5.1. Let $\varphi \in \mathcal{A}^m$ be an automorphism. It is said that $\bar{x} \in S$ is a

- (1) **(local)** strong or strict optimal solution for (23) if there does not exist $(x \in N(\bar{x}, \delta) \cap S \setminus \{\bar{x}\})$ $x \in S \setminus \{\bar{x}\}$ such that $\tilde{F}(x) \tilde{\leq}_\varphi \tilde{F}(\bar{x})$.
- (2) **(local)** optimal solution for (23) if there does not exist $(x \in N(\bar{x}, \delta) \cap S \setminus \{\bar{x}\})$ $x \in S \setminus \{\bar{x}\}$ such that $\tilde{F}(x) \tilde{<}_\varphi \tilde{F}(\bar{x})$.
- (3) **(local)** weak optimal solution for (23) if there does not exist $(x \in N(\bar{x}, \delta) \cap S \setminus \{\bar{x}\})$ $x \in S \setminus \{\bar{x}\}$ such that $\tilde{F}(x) \tilde{<}_\varphi \tilde{F}(\bar{x})$.

The case of multiobjective fuzzy interval maximization problem denoted by $(2MFIO\varphi)$ and called φ -multiobjective fuzzy interval maximization problem

$$(2MFIO\varphi) \max_{x \in S} \tilde{F}(x), \quad (24)$$

is defined as follows.

Definition 5.2. Let $\varphi \in \mathcal{A}^m$ be an automorphism. It is said that $\bar{x} \in S$ is a

- (1) **(local)** strong or strict optimal solution for (24) if there does not exist $(x \in N(\bar{x}, \delta) \cap S \setminus \{\bar{x}\})$ $x \in S \setminus \{\bar{x}\}$ such that $\tilde{F}(\bar{x}) \tilde{\leq}_\varphi \tilde{F}(x)$.
- (2) **(local)** optimal solution for (24) if there does not exist $(x \in N(\bar{x}, \delta) \cap S \setminus \{\bar{x}\})$ $x \in S \setminus \{\bar{x}\}$ such that $\tilde{F}(\bar{x}) \tilde{<}_\varphi \tilde{F}(x)$.
- (3) **(local)** weak optimal solution for (24) if there does not exist $(x \in N(\bar{x}, \delta) \cap S \setminus \{\bar{x}\})$ $x \in S \setminus \{\bar{x}\}$ such that $\tilde{F}(\bar{x}) \tilde{<}_\varphi \tilde{F}(x)$.

Some formulations and concepts of solution for multiobjective fuzzy interval optimization problems given in the literature can be seen as a particular case of the φ -multiobjective fuzzy interval optimization problems as illustrated below.

Example 5.1. Consider $\varphi \in \mathcal{A}^m$ given in Example 2.2, and the preference order relation $\tilde{\leq}_\varphi$ given in Example 4.2 and $S = \{x \in \mathbb{R}^n : g_j(x) \leq 0 \text{ for all } j \in \{1, \dots, k\}\}$ with $g_j : \mathbb{R}^n \rightarrow \mathbb{R}$. Then $(1MFIO\varphi)$ coincides with the formulation of the multiobjective fuzzy interval minimization problem given in [32]. Moreover, the concept of optimal solution for $(1MFIO\varphi)$ coincides with the concept of optimal solution given by Definition 3.1 in [32].

Example 5.2. Consider $\varphi \in \mathcal{A}^m$ given in Example 2.3, and the preference order relation $\tilde{\leq}_\varphi$ given in Example 4.4 and $S = \{x \in \mathbb{R}^n : g_j(x) \leq 0 \text{ for all } j \in \{1, \dots, k\}\}$ with $g_j : \mathbb{R}^n \rightarrow \mathbb{R}$. Then $(1MFIO\varphi)$ coincides with the formulation of the multiobjective fuzzy interval minimization problem given in [19]. Moreover, the concept of optimal solution for $(1MFIO\varphi)$ coincides with the concept of LS-satisficing solution given by Definition 4.3 in [19].

From (18), (19), (20), Definitions 3.1 and 5.1, and from the characterization of a fuzzy interval through its α -levels, φ -multiobjective fuzzy interval optimization problems can be associated with a family of φ -multiobjective interval optimization problems as follows.

Theorem 5.1. Given $\varphi \in \mathcal{A}^m$, consider the φ -multiobjective fuzzy interval optimization problem $(1MFIO\varphi)$ and the φ -multiobjective interval optimization problems $(1MIO\varphi)$ given by

$$(1MIO\varphi) \min_{x \in S} [\tilde{F}]^\alpha(x) \quad (25)$$

for each $\alpha \in [0, 1]$. Then

- (i) $\bar{x} \in S$ is a **(local)** strong or strict optimal solution for (23) if and only if $\bar{x} \in S$ is a **(local)** strong or strict optimal solution for (25) for all $\alpha \in [0, 1]$.
- (ii) $\bar{x} \in S$ is a **(local)** optimal solution for (23) if and only if $\bar{x} \in S$ is a **(local)** optimal solution for (25) for all $\alpha \in [0, 1]$.
- (iii) $\bar{x} \in S$ is a **(local)** weak optimal solution for (23) if and only if $\bar{x} \in S$ is a **(local)** weak optimal solution for (25) for all $\alpha \in [0, 1]$.

5.1. Optimal conditions for multiobjective fuzzy interval optimization problems with preference ordering relation

From Theorem 5.1, the optimality conditions given for multiobjective interval optimization problems presented in Section 3 can be easily extended for φ -multiobjective fuzzy interval optimization problems as illustrated below.

Example 5.3 (Fuzzy version of Example 3.4). Given $\mathbf{0} = (0, \dots, 0) \in \mathbb{R}^n$, consider $(1MFIOP_\varphi)$ such that φ is the automorphism given in Example 2.2, $S = \{x \in \mathbb{R}^n : g_j(x) \leq 0 \text{ for all } j \in \{1, \dots, k\}\}$ with $g_j : \mathbb{R}^n \rightarrow \mathbb{R}$ convex on $\mathbb{R}^n$ and differentiable at $\bar{x}$ for all $j \in \{1, \dots, k\}$, $\bar{x} \in S$ and $\tilde{F} : S \subseteq \mathbb{R}^n \rightarrow (\mathcal{F}_C(\mathbb{R}))^m$ is given by $\tilde{F}(x) = (\tilde{F}_1(x), \dots, \tilde{F}_m(x))$, with $[\tilde{F}_i(x)]^\alpha(x) = [\underline{f}_{i,\alpha}(x), \overline{f}_{i,\alpha}(x)]$, where $\underline{f}_{i,\alpha}, \overline{f}_{i,\alpha} : S \rightarrow \mathbb{R}$ are continuously differentiable at $\bar{x}$ for all $i \in \{1, \dots, m\}$. If $\tilde{F}$ is φ -convex at $\bar{x}$ and if for each $\alpha \in [0, 1]$ there exist $w_1^{(\alpha)}, \dots, w_{2m}^{(\alpha)} \in \mathbb{R}$ with $w_i^{(\alpha)} > 0$ for all $i \in \{1, \dots, 2m\}$ and $\mu_1^{(\alpha)}, \dots, \mu_k^{(\alpha)} \in \mathbb{R}$ with $\mu_j^{(\alpha)} \geq 0$ for all $j \in \{1, \dots, k\}$ such that

$$(i) \sum_{i=1}^m w_{2i-1}^{(\alpha)} \nabla \underline{f}_{i,\alpha}(\bar{x}) + \sum_{i=1}^m w_{2i}^{(\alpha)} \nabla \overline{f}_{i,\alpha}(\bar{x}) + \sum_{j=1}^k \mu_j^{(\alpha)} \nabla g_j(\bar{x}) = \mathbf{0};$$

$$(ii) \mu_j^{(\alpha)} g_j(\bar{x}) = 0 \text{ for all } j \in \{1, \dots, k\},$$

then $\bar{x}$ is an optimal solution for $(1MFIOP_\varphi)$.

Indeed, from φ -convexity of $\tilde{F}$ and from Corollary 4.1, it follows that $[\tilde{F}]^\alpha : S \rightarrow (\mathcal{K}_C)^m$ is φ -convexity for all $\alpha \in [0, 1]$. Thus, for all $\alpha \in [0, 1]$, it follows that $[\tilde{F}]^\alpha$ is a multi-interval-valued function that has all properties required by theorem described in Example 3.4 and, consequently, $\bar{x}$ is an optimal solution for

$$(1MFIOP_\varphi) \min_{x \in S} [\tilde{F}]^\alpha(x)$$

for all $\alpha \in [0, 1]$. Hence, from Theorem 5.1, it follows that $\bar{x}$ is an optimal solution for $(1MFIOP_\varphi)$.

Similar argumentations as those used in Example 5.3 show that the optimality conditions given in the next two examples are obtained.

Example 5.4 (Fuzzy version of Example 3.5). Given $\mathbf{0} = (0, \dots, 0) \in \mathbb{R}^n$, consider $(1MFIOP_\varphi)$ such that φ is the automorphism given in Example 2.4, $S = \{x \in \mathbb{R}^n : g_j(x) \leq 0 \text{ for all } j \in \{1, \dots, k\}\}$ with $g_j : \mathbb{R}^n \rightarrow \mathbb{R}$ convex on $\mathbb{R}^n$ and differentiable at $\bar{x}$ for all $j \in \{1, \dots, k\}$, $\bar{x} \in S$ and $\tilde{F} : S \subseteq \mathbb{R}^n \rightarrow (\mathcal{F}_C(\mathbb{R}))^m$ given by $\tilde{F}(x) = (\tilde{F}_1(x), \dots, \tilde{F}_m(x))$, with $[\tilde{F}_i(x)]^\alpha(x) = [\underline{f}_{i,\alpha}(x), \overline{f}_{i,\alpha}(x)]$, where $\underline{f}_{i,\alpha}, \overline{f}_{i,\alpha} : S \rightarrow \mathbb{R}$ are continuously differentiable at $\bar{x}$ for all $i \in \{1, \dots, m\}$. If $\tilde{F}$ is φ -convex at $\bar{x}$ and if for each $\alpha \in [0, 1]$ there exist $w_1^{(\alpha)}, \dots, w_{2m}^{(\alpha)} \in \mathbb{R}$ with $w_i^{(\alpha)} > 0$ for all $i \in \{1, \dots, 2m\}$ and $\mu_1^{(\alpha)}, \dots, \mu_k^{(\alpha)} \in \mathbb{R}$ with $\mu_j^{(\alpha)} \geq 0$ for all $j \in \{1, \dots, k\}$ such that

$$(i) \sum_{i=1}^m w_{2i-1}^{(\alpha)} \nabla \left(\frac{\underline{f}_{i,\alpha}(\bar{x}) + \overline{f}_{i,\alpha}(\bar{x})}{2} \right) + \sum_{i=1}^m w_{2i}^{(\alpha)} \nabla \left(\frac{\overline{f}_{i,\alpha}(\bar{x}) - \underline{f}_{i,\alpha}(\bar{x})}{2} \right) + \sum_{j=1}^k \mu_j^{(\alpha)} \nabla g_j(\bar{x}) = \mathbf{0};$$

$$(ii) \mu_j^{(\alpha)} g_j(\bar{x}) = 0 \text{ for all } j \in \{1, \dots, k\},$$

then $\bar{x}$ is an optimal solution for $(1MFIOP_\varphi)$.

Example 5.5 (Fuzzy version of Example 3.6). Given $\mathbf{0} = (0, \dots, 0) \in \mathbb{R}^n$, consider $(1MFIOP_\varphi)$ such that φ is the automorphism given in Example 2.4,

$$S = \{x \in \mathbb{R}^n : g_j(x) \leq 0 \text{ for all } j \in \{1, \dots, k\}\}$$

with $g_j : \mathbb{R}^n \rightarrow \mathbb{R}$ convex on $\mathbb{R}^n$ and differentiable at $\bar{x}$ for all $j \in \{1, \dots, k\}$, $\bar{x} \in S$ and $\tilde{F} : S \subseteq \mathbb{R}^n \rightarrow (\mathcal{F}_C(\mathbb{R}))^m$ given by $\tilde{F}(x) = (\tilde{F}_1(x), \dots, \tilde{F}_m(x))$, with $[\tilde{F}_i(x)]^\alpha(x) = [\underline{f}_{i,\alpha}(x), \overline{f}_{i,\alpha}(x)]$, where $\underline{f}_{i,\alpha}, \overline{f}_{i,\alpha} : S \rightarrow \mathbb{R}$ are continuously differentiable at $\bar{x}$ for all $i \in \{1, \dots, m\}$. If $\tilde{F}$ is φ -convex at $\bar{x}$ and if for each $\alpha \in [0, 1]$ there exist $w_1^{(\alpha)}, \dots, w_{2m}^{(\alpha)} \in \mathbb{R}$ with $w_i^{(\alpha)} > 0$ for all $i \in \{1, \dots, 2m\}$ and $\mu_1^{(\alpha)}, \dots, \mu_k^{(\alpha)} \in \mathbb{R}$ with $\mu_j^{(\alpha)} \geq 0$ for all $j \in \{1, \dots, k\}$ such that

$$(i) w_{2i-1}^{(\alpha)} < w_{2i}^{(\alpha)} \text{ for all } i \in \{1, \dots, m\};$$

$$(ii) \sum_{i=1}^m w_{2i-1}^{(\alpha)} \nabla \underline{f}_{i,\alpha}(\bar{x}) + \sum_{i=1}^m w_{2i}^{(\alpha)} \nabla \overline{f}_{i,\alpha}(\bar{x}) + \sum_{j=1}^k \mu_j^{(\alpha)} \nabla g_j(\bar{x}) = \mathbf{0};$$

$$(iii) \mu_j^{(\alpha)} g_j(\bar{x}) = 0 \text{ for all } j \in \{1, \dots, k\},$$

then $\bar{x}$ is an optimal solution for $(1MFIOP_\varphi)$.

Some optimality conditions for multiobjective interval optimization problems presented in the literature, where the use of preference order relations is adopted, can be obtained using the mathematical tools herein developed as illustrated below.

Example 5.6 (Theorem 4.2 in [32]). Given $\mathbf{0} = (0, \dots, 0) \in \mathbb{R}^n$, consider $(1MFIOP_\varphi)$ such that φ is the automorphism given in Example 2.2,

$$S = \{x \in \mathbb{R}^n : g_j(x) \leq 0 \text{ for all } j \in \{1, \dots, k\}\}$$

with $g_j : \mathbb{R}^n \rightarrow \mathbb{R}$ convex on $\mathbb{R}^n$ and continuously differentiable at $\bar{x}$ for all $j \in \{1, \dots, k\}$, $\bar{x} \in S$ and $\tilde{F} : S \subseteq \mathbb{R}^n \rightarrow (\mathcal{F}_C(\mathbb{R}))^m$ given by $\tilde{F}(x) = (\tilde{F}_1(x), \dots, \tilde{F}_m(x))$, with $[\tilde{F}_i(x)]^\alpha(x) = [\underline{f}_{i,\alpha}(x), \bar{f}_{i,\alpha}(x)]$, where $\underline{f}_{i,\alpha}, \bar{f}_{i,\alpha} : S \rightarrow \mathbb{R}$ are continuously differentiable at $\bar{x}$ for all $i \in \{1, \dots, m\}$. If $\tilde{F}$ is φ -convex at $\bar{x}$ and there exist $w_1, \dots, w_{2m} \in \mathbb{R}$ with $w_i > 0$ for all $i \in \{1, \dots, 2m\}$ and $\mu_1, \dots, \mu_k \in \mathbb{R}$ with $\mu_j \geq 0$ for all $j \in \{1, \dots, k\}$ such that

$$\begin{aligned} \text{(i)} \quad & \sum_{i=1}^m w_{2i-1} \int_0^1 \nabla \underline{f}_{i,\alpha}(\bar{x}) d\alpha + \sum_{i=1}^m w_{2i} \int_0^1 \nabla \bar{f}_{i,\alpha}(\bar{x}) d\alpha + \sum_{j=1}^k \mu_j \nabla g_j(\bar{x}) = \mathbf{0}; \\ \text{(ii)} \quad & \mu_j g_j(\bar{x}) = 0 \text{ for all } j \in \{1, \dots, k\}, \end{aligned}$$

then $\bar{x}$ is an optimal solution for $(1MFIOP_\varphi)$.

Next, it is shown that solving $(1MFIOP_\varphi)$ and $(2MFIOP_\varphi)$ is equivalent solving a family of classical multiobjective optimization problems of the type (12). To this end, consider the family of maps

$$(\tilde{\varphi} \circ [\tilde{F}]^\alpha) : S \rightarrow \mathbb{R}^{2m}$$

given by

$$\begin{aligned} (\tilde{\varphi} \circ [\tilde{F}]^\alpha)(x) &= \tilde{\varphi}([\tilde{F}(x)]^\alpha) \\ &= \tilde{\varphi}([\tilde{F}_1(x)]^\alpha, \dots, [\tilde{F}_m(x)]^\alpha) = \tilde{\varphi}([\underline{f}_{1,\alpha}(x), \bar{f}_{1,\alpha}(x)], \dots, [\underline{f}_{m,\alpha}(x), \bar{f}_{m,\alpha}(x)]) \\ &= \left(\varphi_1(\underline{f}_{1,\alpha}(x), \bar{f}_{1,\alpha}(x)), \dots, \varphi_m(\underline{f}_{m,\alpha}(x), \bar{f}_{m,\alpha}(x)) \right) \\ &= \left((\lambda_1 \underline{f}_{1,\alpha} + \lambda_2 \bar{f}_{1,\alpha})(x), (\beta_1 \underline{f}_{1,\alpha} + \beta_2 \bar{f}_{1,\alpha})(x), \dots, (\lambda_{2m-1} \underline{f}_{m,\alpha} + \lambda_{2m} \bar{f}_{m,\alpha})(x), (\beta_{2m-1} \underline{f}_{m,\alpha} + \beta_{2m} \bar{f}_{m,\alpha})(x) \right) \\ &= \left(\Lambda_1^T f_\alpha(x), B_1^T f_\alpha(x), \dots, \Lambda_{2m}^T f_\alpha(x), B_{2m}^T f_\alpha(x) \right), \end{aligned}$$

where $\varphi \in \mathcal{A}^m$ and $f_\alpha : S \rightarrow \mathbb{R}^{2m}$ is defined by $f_\alpha(x) = (\underline{f}_{1,\alpha}(x), \bar{f}_{1,\alpha}(x), \dots, \underline{f}_{m,\alpha}(x), \bar{f}_{m,\alpha}(x))$ for all $x \in S$ and for each $\alpha \in [0, 1]$. Thus, associated with the φ -multiobjective fuzzy interval optimization problems (23) and (24) the following family of classical multiobjective minimization problems

$$(MOP_{\varphi,\alpha}) \min_{x \in S} (\tilde{\varphi} \circ [F]^\alpha)(x) \quad (26)$$

is obtained. Using the usual partial order relation (2) in $\mathbb{R}^{2m}$ and strict partial order relations (4) and (5), the usual solution (efficient and weakly efficient) concept for (26) is obtained.

Theorem 5.2. $\bar{x} \in S$ is an optimal solution (weakly optimal solution) for $(1MFIOP_\varphi)$ if and only if $\bar{x} \in S$ is an efficient solution (weakly efficient solution) for $(MOP_{\varphi,\alpha})$ for all $\alpha \in [0, 1]$.

Proof. See the proof in Appendix A. $\square$

Theorem 5.3. $\bar{x} \in S$ is an optimal solution (weak optimal solution) for $(2MFIOP_\varphi)$ if and only if $\bar{x} \in S$ is an efficient solution (weakly efficient solution) for $(MOP_{(-\varphi),\alpha})$ for all $\alpha \in [0, 1]$.

Proof. See the proof in Appendix A. $\square$

Theorems 5.2 and 5.3 demonstrate that to solve the problems (23) and (24), the usual multiobjective programming techniques can be used. In particular, such as scalarization, or interactive methods can be used along with the associated condition of optimality type Karush-Kuhn-Tucker. This is illustrated below.

Example 5.7 (Scalarization 2). Associated with (26), consider the weighted scalar problem (see [7])

$$(MOP_{\varphi,\alpha}(w^{(\alpha)})) \min_{x \in S} \sum_{i=1}^m (w_{2i-1}^{(\alpha)} \Lambda_i^T f(x) + w_{2i}^{(\alpha)} B_i^T f(x)), \quad (27)$$

where $w^{(\alpha)} = (w_1^{(\alpha)}, \dots, w_{2m}^{(\alpha)}) \in \mathbb{R}^{2m}$, $w_i^{(\alpha)} \geq 0$ for all $i \in \{1, \dots, 2m\}$ and $\sum_{i=1}^{2m} w_i^{(\alpha)} = 1$. The relationships between the solutions of the problems (23) and (24) and the solutions of the scalar problem (27) are as follows

1. If $\bar{x} \in S$ is an optimal solution for $(MOP_{\varphi, \alpha}(w^{(\alpha)}))$ for all $\alpha \in [0, 1]$, then $\bar{x} \in S$ is a weak optimal solution for $(1MFIOP_{\varphi})$.
2. If $\bar{x} \in S$ is an optimal solution for $(MOP_{\varphi, \alpha}(w^{(\alpha)}))$, with $w_i^{(\alpha)} > 0$ for all $i \in \{1, \dots, 2m\}$ and for all $\alpha \in [0, 1]$, then $\bar{x} \in S$ is an optimal solution for $(1MFIOP_{\varphi})$.
3. If $\bar{x} \in S$ is the only optimal solution for each $(MOP_{\varphi, \alpha}(w^{(\alpha)}))$, $\alpha \in [0, 1]$, then $\bar{x} \in S$ is an optimal solution for $(1MFIOP_{\varphi})$.
4. If S is convex, F is φ -convex and $\bar{x} \in S$ is an optimal solution for $(1MFIOP_{\varphi})$, then for all $\alpha \in [0, 1]$ there exists

$$w^{(\alpha)} = (w_1^{(\alpha)}, \dots, w_{2m}^{(\alpha)}) \in \mathbb{R}^{2m} \text{ with } w_i^{(\alpha)} \geq 0 \text{ for all } i \in \{1, \dots, 2m\} \text{ and } \sum_{i=1}^{2m} w_i^{(\alpha)} = 1 \text{ such that } \bar{x} \in S \text{ is an optimal solution for } (MOP_{\varphi, \alpha}(w^{(\alpha)})).$$

Example 5.8 (Necessary and sufficient optimality conditions 2). Given $\tilde{F} : S \subseteq \mathbb{R}^n \rightarrow (\mathcal{F}_C(\mathbb{R}))^m$ such that $\tilde{F}(x) = (\tilde{F}_1(x), \dots, \tilde{F}_m(x))$, with $[\tilde{F}_i(x)]^{\alpha}(x) = [\underline{f}_{i, \alpha}(x), \bar{f}_{i, \alpha}(x)]$ for all $i \in \{1, \dots, m\}$ and for all $\alpha \in [0, 1]$, let $\Lambda_i^T f$, $B_i^T f$ be differentiable for all $i \in \{1, \dots, m\}$ and for all $\alpha \in [0, 1]$ and consider $\mathbf{0} = (0, \dots, 0) \in \mathbb{R}^n$. Then

1. If $\bar{x} \in S$ is a weak optimal solution for $(1MFIOP_{\varphi})$, then for each $\alpha \in [0, 1]$ there are $w_1^{(\alpha)}, \dots, w_{2m}^{(\alpha)} \in \mathbb{R}$, with $w_i^{(\alpha)} \geq 0$ not equal zero for all $i \in \{1, \dots, 2m\}$, such that

$$\sum_{i=1}^m w_{2i-1}^{(\alpha)} \nabla \Lambda_i^T f_{\alpha}(\bar{x}) + \sum_{i=1}^m w_{2i}^{(\alpha)} \nabla B_i^T f_{\alpha}(\bar{x}) = \sum_{i=1}^m \nabla \left[\left(w_{2i-1}^{(\alpha)} \Lambda_i^T + w_{2i}^{(\alpha)} B_i^T \right) f_{\alpha} \right](\bar{x}) = \mathbf{0}. \quad (28)$$

2. If $\bar{x} \in S$, S is convex, F is φ -convex and for each $\alpha \in [0, 1]$ there are $w_1^{(\alpha)}, \dots, w_{2m}^{(\alpha)} \in \mathbb{R}$ with $w_i^{(\alpha)} \geq 0$ not equal zero for all $i \in \{1, \dots, 2m\}$, such that (28) holds, then $\bar{x}$ is a weak optimal solution for $(1MFIOP_{\varphi})$.
3. If $\bar{x} \in S$, S is convex, F is φ -convex and there are for each $\alpha \in [0, 1]$ there are $w_1^{(\alpha)}, \dots, w_{2m}^{(\alpha)} \in \mathbb{R}$ with $w_i^{(\alpha)} > 0$ for all $i \in \{1, \dots, 2m\}$, such that (28) holds, then $\bar{x}$ is an optimal solution for $(1MFIOP_{\varphi})$.

There exists a semantic meaning behind of each φ -multiobjective fuzzy interval problem, that is interpreted by means of its concept of solution, which is given by means of an automorphism $\varphi \in \mathcal{A}^m$. Consequently, for each $\varphi \in \mathcal{A}^m$, there exists a semantic meaning for the associated φ -multiobjective fuzzy interval problem. However, there exist cases in which it is possible to obtain optimality conditions for a φ -multiobjective fuzzy interval problem from the optimality conditions for another one as illustrated in the next result.

Proposition 5.1. Let $(1MFIOP_{\varphi})$ and $(1MFIOP_{\psi})$ be the multiobjective fuzzy interval optimization problems such that φ is the automorphism given in Example 2.2 and $\psi = (\psi_1 \times \dots \times \psi_m) \in \mathcal{A}^m$ the automorphism given in Example 2.3, i.e.,

$$\psi_i(x_{2i-1}, x_{2i}) = (x_{2i-1}, x_{2i} - x_{2i-1})$$

for all $i \in \{1, \dots, m\}$. If $\bar{x} \in S$ is an optimal solution for $(1MFIOP_{\varphi})$, then $\bar{x}$ is also an optimal solution for $(1MFIOP_{\psi})$.

Proof. See the proof in Appendix A. $\square$

Remark 5.1. From Proposition 5.1 each optimal condition for $(1MFIOP_{\varphi})$, where φ is the automorphism given in Example 2.2, it is also an optimal condition for $(1MFIOP_{\psi})$, where $\psi = (\psi_1 \times \dots \times \psi_m) \in \mathcal{A}^m$ is the automorphism given in Example 2.3, i.e.,

$$\psi_i(x_{2i-1}, x_{2i}) = (x_{2i-1}, x_{2i} - x_{2i-1})$$

for all $i \in \{1, \dots, m\}$. Thus, it is possible to obtain optimal conditions for $(1MFIOP_{\psi})$ by means of the optimal conditions obtained for $(1MFIOP_{\varphi})$.

6. Conclusion

The mathematical tools presented in this study provide, through the preference order relation $\leq_{\varphi}$, the ability to generate order relations on the space of all m -tuples of intervals by choosing appropriate automorphisms φ . Using these order relations it is possible to provide the formulations of the multiobjective interval minimization and maximization problems, $(1MIOP_{\varphi})$ and $(2MIOP_{\varphi})$, that may better represent the phenomena analyzed. For example, it is well-known that the interval space does not have a structure of linear (vectorial) space. However, there exist different ways of providing a structure of quasilinear space to it depending on the chosen partial order relation (see, e.g., [23]). Thus, considering a simple case, where $m = 1$, if one of the objectives of a decision making is

to model an interval optimization problem aiming to use properties of an ordered quasilinear interval space, then semantically the use of the partial interval order $\leq_{\varphi_1}$, where φ_1 is given in Example 2.2, it is not suitable since the interval space equipped with this order does not satisfy the axioms of an ordered quasilinear interval space. To be more precise, the interval space equipped with this partial order does not satisfy the axiom (q.11) of Definition 2.1 in [23] and, in order to verify this fact one can consider $\mathcal{X} = \mathcal{K}_C$, $\alpha = 1$, $\beta = -1$ and $x = [-1, 1]$. On the other hand, it is well-known that the preference partial order relation $\leq_{\varphi_2}$, where $\varphi_2 \in \mathcal{A}$ is given by $\varphi_2(x_1, x_2) = (\lambda_1 x_1 + \lambda_2 x_2, \beta_1 x_1 + \beta_2 x_2)$, with $\lambda_1 = -1$, $\lambda_2 = 0 = \beta_1$, and $\beta_2 = 1$, and which coincides with the inclusion order $\subseteq$, provides a structure of quasilinear space for the interval space (see, e.g., [23]). Consequently, the use of $\leq_{\varphi_2}$ would be a better option than the use of $\leq_{\varphi_1}$ in a decision making. Another advantage in using the preference order relations $\leq_{\varphi}$ is that, to solve $(1MIOp_{\varphi})$ and $(2MIOp_{\varphi})$ we can use all the well-known tools developed to solve with classical multiobjective optimization problems, since they are equivalent, as shown in Theorem 3.1, Theorem 3.2 and Examples 3.8 and 3.9.

The results obtained in the study of $(1MIOp_{\varphi})$ and $(2MIOp_{\varphi})$ not only extend those presented in [9], but it also can be applied to some multiobjective interval-valued optimization problems presented in the literature. For example, considering $\varphi \in \mathcal{A}^m$ given in Example 2.2, it follows that $(1MIOp_{\varphi})$ and its concept of solution coincide with the formulation of multiobjective fuzzy interval minimization problem and one of its solutions types (type-I Pareto), respectively, given in [31]. On the other hand, considering $\varphi \in \mathcal{A}^m$ given in Example 2.4, it follows that $(1MIOp_{\varphi})$ and its concept of solution coincide with the formulation of multiobjective fuzzy interval minimization problem and another of its solutions types (type-II Pareto), respectively, given in [31]. Now, considering $\varphi \in \mathcal{A}^m$ given in Example 2.3, it follows that $(1MIOp_{\varphi})$ and its concept of solution coincide with the formulation of multiobjective fuzzy interval minimization problem and one of its solutions types (LS-Pareto), respectively, given in [26].

The α -levels of a m -tuple of fuzzy intervals $\tilde{u} = (\tilde{u}_1, \dots, \tilde{u}_m) \in (\mathcal{F}_C(\mathbb{R}))^m$ are m -tuples of intervals $([\tilde{u}_1]^\alpha, \dots, [\tilde{u}_m]^\alpha) \in (\mathcal{K}_C)^m$ for all $\alpha \in [0, 1]$ (see, e.g., [13]). Therefore, it follows that the order relation $\leq_{\varphi}$ can be extended into space of all m -tuple of fuzzy intervals $(\mathcal{F}_C(\mathbb{R}))^m$ defining the binary relation $\tilde{\leq}_{\varphi}$ on $(\mathcal{F}_C(\mathbb{R}))^m \times (\mathcal{F}_C(\mathbb{R}))^m$ by $\tilde{u} \tilde{\leq}_{\varphi} \tilde{v} \Leftrightarrow ([\tilde{u}_1]^\alpha, \dots, [\tilde{u}_m]^\alpha) \leq_{\varphi} ([\tilde{v}_1]^\alpha, \dots, [\tilde{v}_m]^\alpha)$ for all $\alpha \in [0, 1]$. Thus, $\tilde{\leq}_{\varphi}$ is a partial order relation on fuzzy-interval space and, consequently, the formulations for φ -multiobjective fuzzy interval optimization problems $(1MFIOP_{\varphi})$ and $(2MFIOP_{\varphi})$, can be obtained. To solve $(1MFIOP_{\varphi})$ and $(2MFIOP_{\varphi})$ we can use all the well-known tools developed to solve multiobjective optimization problems, as shown in Theorem 5.2, Theorem 5.3 and Examples 5.7 and 5.8.

Some formulations of solutions concepts for multiobjective fuzzy interval optimization problems given in the literature can be seen as a particular case of a $(1MFIOP_{\varphi})$, where $\varphi \in \mathcal{A}^m$. For example, considering $\varphi \in \mathcal{A}^m$ given in Example 2.2, it follows that $(1MFIOP_{\varphi})$ and its concept of solution coincide with the formulation of multiobjective fuzzy interval minimization problem and its concept of solution, respectively, given in [32]. Now, considering $\varphi \in \mathcal{A}^m$ given in Example 2.3, it follows that $(1MFIOP_{\varphi})$ and its concept of solution coincide with one of the formulations of multiobjective fuzzy interval minimization problem and its concept of solution given in [19]. Moreover, if $m = 1$, then it can be seen $(1MFIOP_{\varphi})$ as a fuzzy interval optimization problem whose objective function is a map of type $\tilde{F} : S \subseteq \mathbb{R}^n \rightarrow \mathcal{F}_C(\mathbb{R})$ and denote $(1MFIOP_{\varphi})$ by $(1FIOp_{\varphi})$. For this case, other formulations of fuzzy interval optimization problems given in the literature also can be seen as a particular case of $(1FIOp_{\varphi})$. For example, considering $\varphi \in \mathcal{A}$ given in Example 2.2, it follows that $(1FIOp_{\varphi})$ and its concept of solution coincide with the formulations of fuzzy interval minimization problem and its concept of solutions given in [27,30]. Therefore, this presentation can be considered as a unified approach to dealing with multiobjective interval and multiobjective fuzzy interval optimization problems that are formulated by means of preference order relations.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Proofs

Proof of Theorem 5.2. Let $\bar{x} \in S$ be is an optimal solution for $(1MFIOP_{\varphi})$. Consequently, from Theorem 5.1 there do not exist $x \in S$ and $\alpha \in [0, 1]$ such that $[\tilde{F}(x)]^\alpha \leq_{\varphi} [\tilde{F}(\bar{x})]^\alpha$. If $\bar{x}$ is not an efficient solution for $(MOP_{\varphi, \alpha})$ for all $\alpha \in [0, 1]$, then there exists $\hat{\alpha} \in [0, 1]$ and $x \in S$, with $x \neq \bar{x}$, such that

$$\Lambda_i^T f_{\hat{\alpha}}(x) \leq \Lambda_i^T f_{\hat{\alpha}}(\bar{x}) \quad \text{and} \quad B_i^T f_{\hat{\alpha}}(x) \leq B_i^T f_{\hat{\alpha}}(\bar{x})$$

for all $i \in \{1, \dots, m\}$ and, consequently,

$$\begin{aligned} & (\varphi_1(f_{\underline{1},\hat{\alpha}}(x), \bar{f}_{1,\hat{\alpha}}(x)), \dots, \varphi_m(f_{\underline{m},\hat{\alpha}}(x), \bar{f}_{m,\hat{\alpha}}(x))) \leq^{2m} (\varphi_1(f_{\underline{1},\hat{\alpha}}(\bar{x}), \bar{f}_{1,\hat{\alpha}}(\bar{x})), \dots, \varphi_m(f_{\underline{m},\hat{\alpha}}(\bar{x}), \bar{f}_{m,\hat{\alpha}}(\bar{x}))) \\ & \Rightarrow \varphi(f_{\underline{1},\hat{\alpha}}(x), \bar{f}_{1,\hat{\alpha}}(x), \dots, f_{\underline{m},\hat{\alpha}}(x), \bar{f}_{m,\hat{\alpha}}(x)) \leq^{2m} \varphi(f_{\underline{1},\hat{\alpha}}(\bar{x}), \bar{f}_{1,\hat{\alpha}}(\bar{x}), \dots, f_{\underline{m},\hat{\alpha}}(\bar{x}), \bar{f}_{m,\hat{\alpha}}(\bar{x})) \\ & \Rightarrow \bar{\varphi}([f_{\underline{1},\hat{\alpha}}(x), \bar{f}_{1,\hat{\alpha}}(x)], \dots, [f_{\underline{m},\hat{\alpha}}(x), \bar{f}_{m,\hat{\alpha}}(x)]) \leq^{2m} \bar{\varphi}([f_{\underline{1},\hat{\alpha}}(\bar{x}), \bar{f}_{1,\hat{\alpha}}(\bar{x})], \dots, [f_{\underline{m},\hat{\alpha}}(\bar{x}), \bar{f}_{m,\hat{\alpha}}(\bar{x})]) \\ & \Rightarrow \bar{\varphi}([\bar{F}(x)]^{\hat{\alpha}}) \leq^{2m} \bar{\varphi}([\bar{F}(\bar{x})]^{\hat{\alpha}}) \Rightarrow [\bar{F}(x)]^{\hat{\alpha}} \leq_{\varphi} [\bar{F}(\bar{x})]^{\hat{\alpha}}. \end{aligned}$$

However, this contradicts the fact that $\bar{x}$ is an optimal solution for $(1MFIO P_{\varphi})$. Therefore, if $\bar{x} \in S$ is an optimal solution for $(1MFIO P_{\varphi})$, then $\bar{x} \in S$ is an efficient solution for $(MOP_{\varphi,\alpha})$ for all $\alpha \in [0, 1]$.

Reciprocally, let $\bar{x} \in S$ be an efficient solution for $(MOP_{\varphi,\alpha})$ for all $\alpha \in [0, 1]$. If $\bar{x}$ is not an optimal solution for $(1MFIO P_{\varphi})$, then from Theorem 5.1 there exist $\hat{\alpha} \in [0, 1]$ and $x \in S$, with $x \neq \bar{x}$, such that $[\bar{F}(x)]^{\hat{\alpha}} \leq_{\varphi} [\bar{F}(\bar{x})]^{\hat{\alpha}}$, that is, $\bar{\varphi}([\bar{F}(x)]^{\hat{\alpha}}) \leq^{2m} \bar{\varphi}([\bar{F}(\bar{x})]^{\hat{\alpha}})$. Thus,

$$\Lambda_i^T f_{\hat{\alpha}}(x) \leq \Lambda_i^T f_{\hat{\alpha}}(\bar{x}) \quad \text{and} \quad B_i^T f_{\hat{\alpha}}(x) \leq B_i^T f_{\hat{\alpha}}(\bar{x})$$

for all $i \in \{1, \dots, m\}$. However, this contradicts the fact that $\bar{x}$ is an efficient solution for $(MOP_{\varphi,\alpha})$ for all $\alpha \in [0, 1]$. Therefore, if $\bar{x} \in S$ is an efficient solution for $(MOP_{\varphi,\alpha})$ for all $\alpha \in [0, 1]$, then $\bar{x} \in S$ is an optimal solution for $(1MFIO P_{\varphi})$.

Using similar arguments, the proof for weak solutions can be obtained. $\square$

Proof of Theorem 5.3. Given $-\varphi \in \mathcal{A}^m$ by

$$\begin{aligned} -\varphi(a_1, \dots, a_{2m}) = & ((-\lambda_1 a_1 - \lambda_2 a_2, -\beta_1 a_1 - \beta_2 a_2), \dots, \\ & \dots, (-\lambda_{2m-1} a_{2m-1} - \lambda_{2m} a_{2m}, -\beta_{2m-1} a_{2m-1} - \beta_{2m} a_{2m})), \end{aligned}$$

consider

$$(MOP_{(-\varphi),\alpha}) \min_{x \in S} (-\Lambda_1^T f_{\alpha}(x), -B_1^T f_{\alpha}(x), \dots, -\Lambda_m^T f_{\alpha}(x), -B_m^T f_{\alpha}(x))$$

for each $\alpha \in [0, 1]$.

Let $\bar{x} \in S$ be an optimal solution for $(2MFIO P_{\varphi})$. Consequently, from Theorem 5.1 there do not exist $x \in S$ and $\alpha \in [0, 1]$ such that $[\bar{F}(\bar{x})]^{\alpha} \leq_{\varphi} [\bar{F}(x)]^{\alpha}$. If $\bar{x}$ is not an efficient solution for $(MOP_{(-\varphi),\alpha})$ for all $\alpha \in [0, 1]$, then there exist $\hat{\alpha} \in [0, 1]$ and $x \in S$, with $x \neq \bar{x}$, such that

$$-\Lambda_i^T f_{\hat{\alpha}}(x) \leq -\Lambda_i^T f_{\hat{\alpha}}(\bar{x}) \quad \text{and} \quad -B_i^T f_{\hat{\alpha}}(x) \leq -B_i^T f_{\hat{\alpha}}(\bar{x}),$$

for all $i \in \{1, \dots, m\}$ and, consequently,

$$\begin{aligned} & (\varphi_1(f_{\underline{1},\hat{\alpha}}(\bar{x}), \bar{f}_{1,\hat{\alpha}}(\bar{x})), \dots, \varphi_m(f_{\underline{m},\hat{\alpha}}(\bar{x}), \bar{f}_{m,\hat{\alpha}}(\bar{x}))) \leq^{2m} (\varphi_1(f_{\underline{1},\hat{\alpha}}(x), \bar{f}_{1,\hat{\alpha}}(x)), \dots, \varphi_m(f_{\underline{m},\hat{\alpha}}(x), \bar{f}_{m,\hat{\alpha}}(x))) \\ & \Rightarrow \varphi(f_{\underline{1},\hat{\alpha}}(\bar{x}), \bar{f}_{1,\hat{\alpha}}(\bar{x}), \dots, f_{\underline{m},\hat{\alpha}}(\bar{x}), \bar{f}_{m,\hat{\alpha}}(\bar{x})) \leq^{2m} \varphi(f_{\underline{1},\hat{\alpha}}(x), \bar{f}_{1,\hat{\alpha}}(x), \dots, f_{\underline{m},\hat{\alpha}}(x), \bar{f}_{m,\hat{\alpha}}(x)) \\ & \Rightarrow \bar{\varphi}([f_{\underline{1},\hat{\alpha}}(\bar{x}), \bar{f}_{1,\hat{\alpha}}(\bar{x})], \dots, [f_{\underline{m},\hat{\alpha}}(\bar{x}), \bar{f}_{m,\hat{\alpha}}(\bar{x})]) \leq^{2m} \bar{\varphi}([f_{\underline{1},\hat{\alpha}}(x), \bar{f}_{1,\hat{\alpha}}(x)], \dots, [f_{\underline{m},\hat{\alpha}}(x), \bar{f}_{m,\hat{\alpha}}(x)]) \\ & \Rightarrow \bar{\varphi}([\bar{F}(\bar{x})]^{\hat{\alpha}}) \leq^{2m} \bar{\varphi}([F(x)]^{\hat{\alpha}}) \Rightarrow [\bar{F}(\bar{x})]^{\hat{\alpha}} \leq_{\varphi} [F(x)]^{\hat{\alpha}}. \end{aligned}$$

However, this contradicts the fact that $\bar{x}$ is an optimal solution for $(2MFIO P_{\varphi})$. Therefore, if $\bar{x} \in S$ is an optimal solution for $(2MFIO P_{\varphi})$, then $\bar{x}$ is an efficient solution for $(MOP_{(-\varphi),\alpha})$ for all $\alpha \in [0, 1]$.

Reciprocally, let $\bar{x} \in S$ be an efficient solution for $(MOP_{(-\varphi),\alpha})$ for $\alpha \in [0, 1]$. If $\bar{x}$ is not an optimal solution for $(2MFIO P_{\varphi})$, then there exist $\hat{\alpha} \in [0, 1]$ and $x \in S$, with $x \neq \bar{x}$, such that $[\bar{F}(\bar{x})]^{\hat{\alpha}} \leq_{\varphi} [F(x)]^{\hat{\alpha}}$ and, consequently, $\bar{\varphi}([\bar{F}(\bar{x})]^{\hat{\alpha}}) \leq^{2m} \bar{\varphi}([F(x)]^{\hat{\alpha}})$. That is,

$$\varphi(f_{\underline{1},\hat{\alpha}}(\bar{x}), \bar{f}_{1,\hat{\alpha}}(\bar{x}), \dots, f_{\underline{m},\hat{\alpha}}(\bar{x}), \bar{f}_{m,\hat{\alpha}}(\bar{x})) \leq^{2m} \varphi(f_{\underline{1},\hat{\alpha}}(x), \bar{f}_{1,\hat{\alpha}}(x), \dots, f_{\underline{m},\hat{\alpha}}(x), \bar{f}_{m,\hat{\alpha}}(x)).$$

Thus,

$$\begin{aligned} \Lambda_i^T f_{\hat{\alpha}}(\bar{x}) & \leq \Lambda_i^T f_{\hat{\alpha}}(x) \quad \text{and} \quad B_i^T f_{\hat{\alpha}}(\bar{x}) \leq B_i^T f_{\hat{\alpha}}(x) \\ -\Lambda_i^T f_{\hat{\alpha}}(x) & \leq -\Lambda_i^T f_{\hat{\alpha}}(\bar{x}) \quad \text{and} \quad -B_i^T f_{\hat{\alpha}}(x) \leq -B_i^T f_{\hat{\alpha}}(\bar{x}). \end{aligned}$$

However, this contradicts the fact that $\bar{x}$ is an efficient solution for $(MOP_{(-\varphi),\alpha})$ for all $\alpha \in [0, 1]$. Therefore, if $\bar{x} \in S$ is an efficient solution for $(MOP_{(-\varphi),\alpha})$ for all $\alpha \in [0, 1]$, then $\bar{x}$ is an optimal solution for $(2MFIO P_{\varphi})$.

Using similar arguments, the proof for weak solutions can be obtained. $\square$

Proof of Proposition 5.1. Let $\bar{x} \in S$ be an optimal solution for $(1MFIOP_\varphi)$. If $\bar{x} \in S$ is not an optimal solution for $(1MFIOP_\psi)$, then there exists $x \in S$ with $x \neq \bar{x}$ such that $\bar{F}(x) \leq_\psi \bar{F}(\bar{x})$. Consequently, there exists $x \in S$ with $x \neq \bar{x}$ such that

$$(I) \quad \underline{f}_{i,\alpha}(x) \leq \underline{f}_{i,\alpha}(\bar{x})$$

and

$$(II) \quad (\bar{f}_{i,\alpha} - \underline{f}_{i,\alpha})(x) \leq (\bar{f}_{i,\alpha} - \underline{f}_{i,\alpha})(\bar{x})$$

for all $\alpha \in [0, 1]$ and for all $i \in \{1, \dots, m\}$. However, from (I) and (II), it follows that there does not exist $x \in S$ with $x \neq \bar{x}$ such that $\underline{f}_{i,\alpha}(x) \leq \underline{f}_{i,\alpha}(\bar{x})$ and $\bar{f}_{i,\alpha}(x) \leq \bar{f}_{i,\alpha}(\bar{x})$ for all $\alpha \in [0, 1]$ and for all $i \in \{1, \dots, m\}$, which is equivalent to the non-existence of $x \neq \bar{x}$ such that

$$[\underline{f}_{i,\alpha}(x), \bar{f}_{i,\alpha}(x)] \leq_{\varphi_i} [\underline{f}_{i,\alpha}(\bar{x}), \bar{f}_{i,\alpha}(\bar{x})]$$

for all $\alpha \in [0, 1]$ and for all $i \in \{1, \dots, m\}$. Thus, from definition of $\leq_\varphi$, Definition 3.1, and from Theorem 5.1, it follows that $\bar{x}$ is not an optimal solution for $(1MFIOP_\varphi)$, which is a contradiction. Therefore, if $\bar{x} \in S$ is an optimal solution for $(1MFIOP_\varphi)$, then $\bar{x}$ is also an optimal solution for $(1MFIOP_\psi)$. $\square$

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